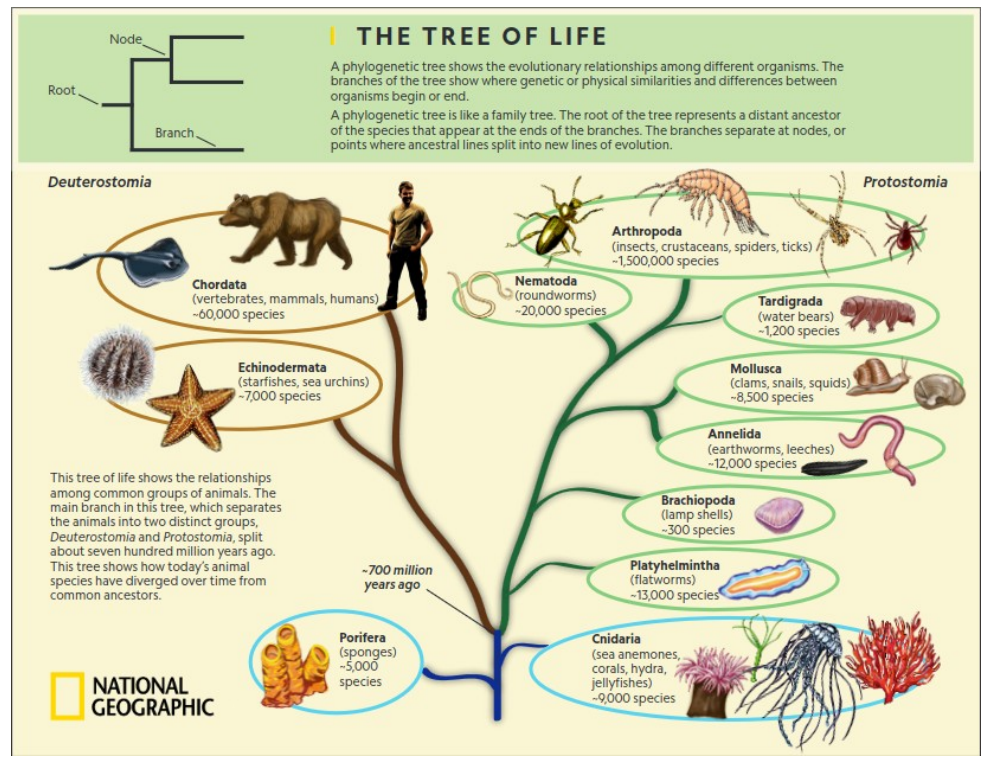
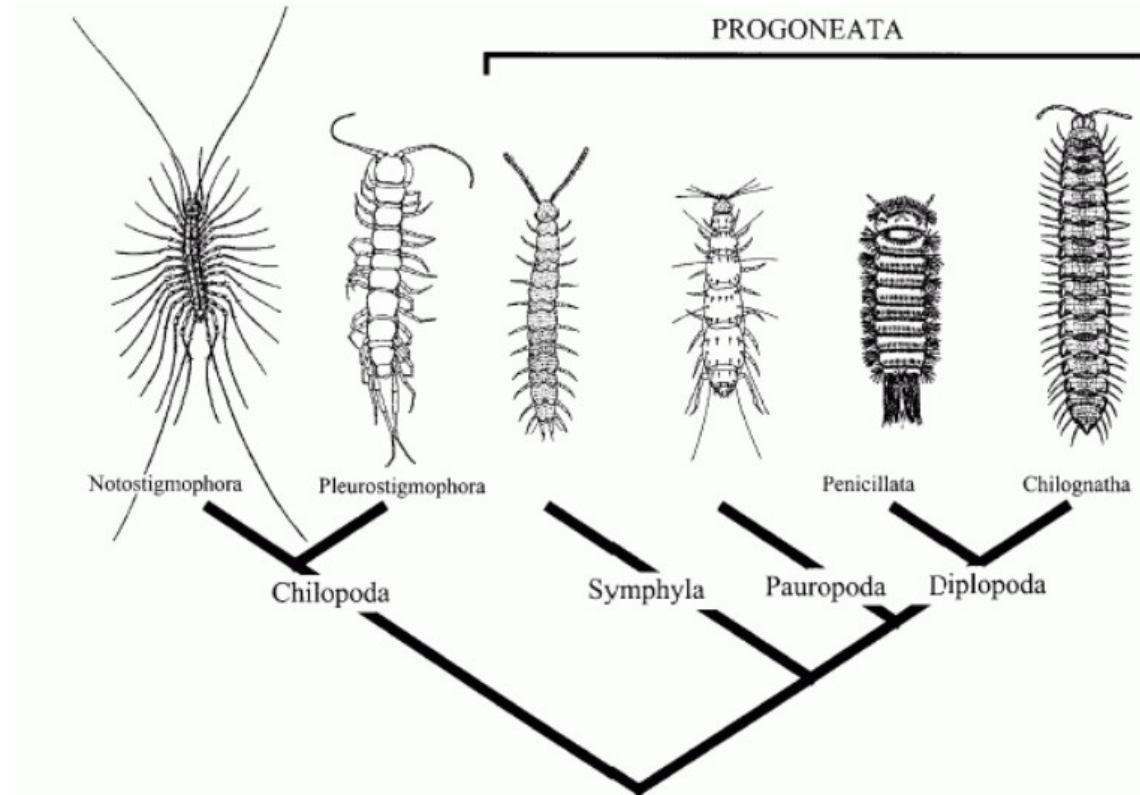


Phylogenetic Analysis

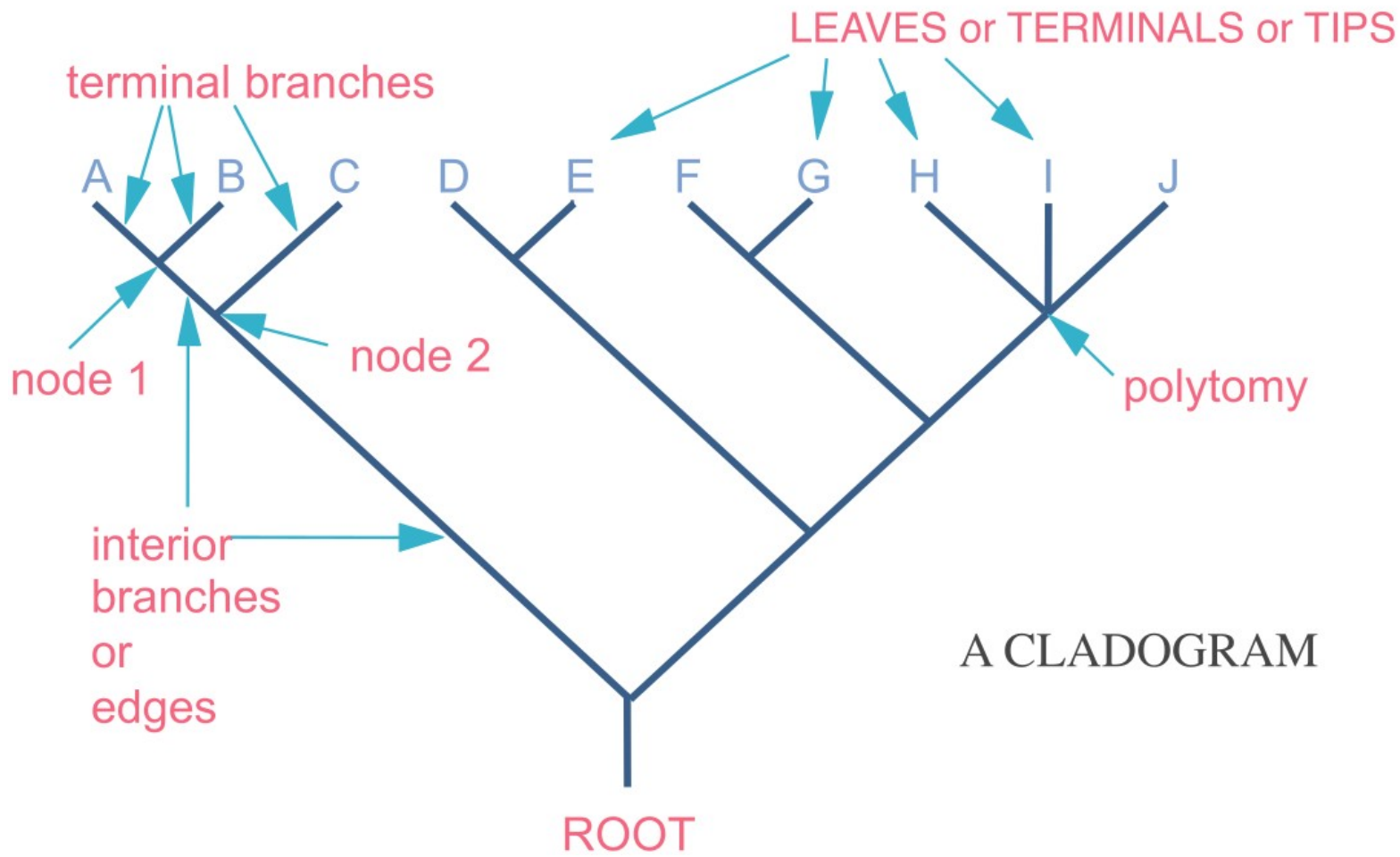
Nikolay Oskolkov, Biology Department, Lund University,
Principal Investigator, Group Leader at NIRI





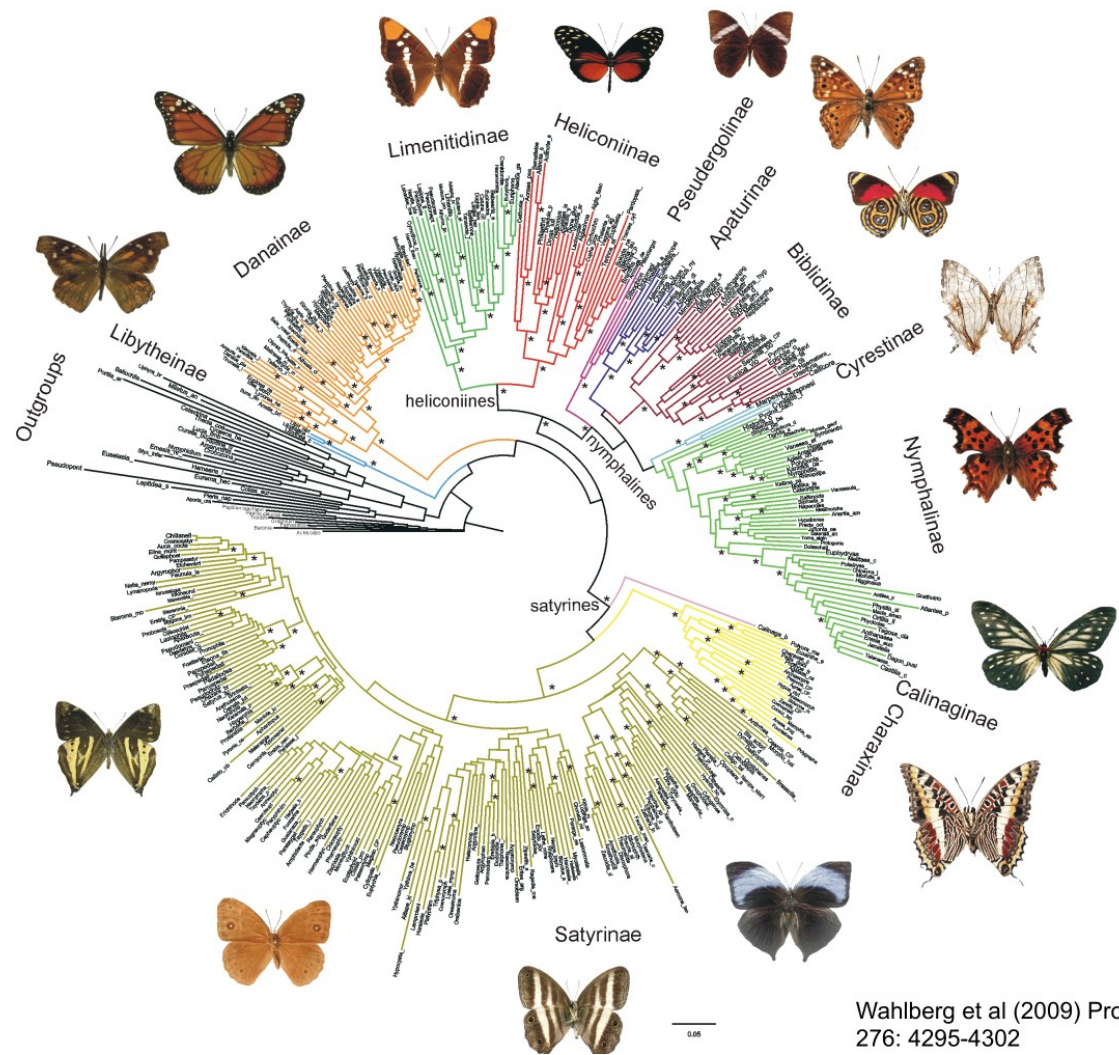
A phylogeny is an inference of the historical genealogy of groups of organisms

Terminology of a phylogenetic tree

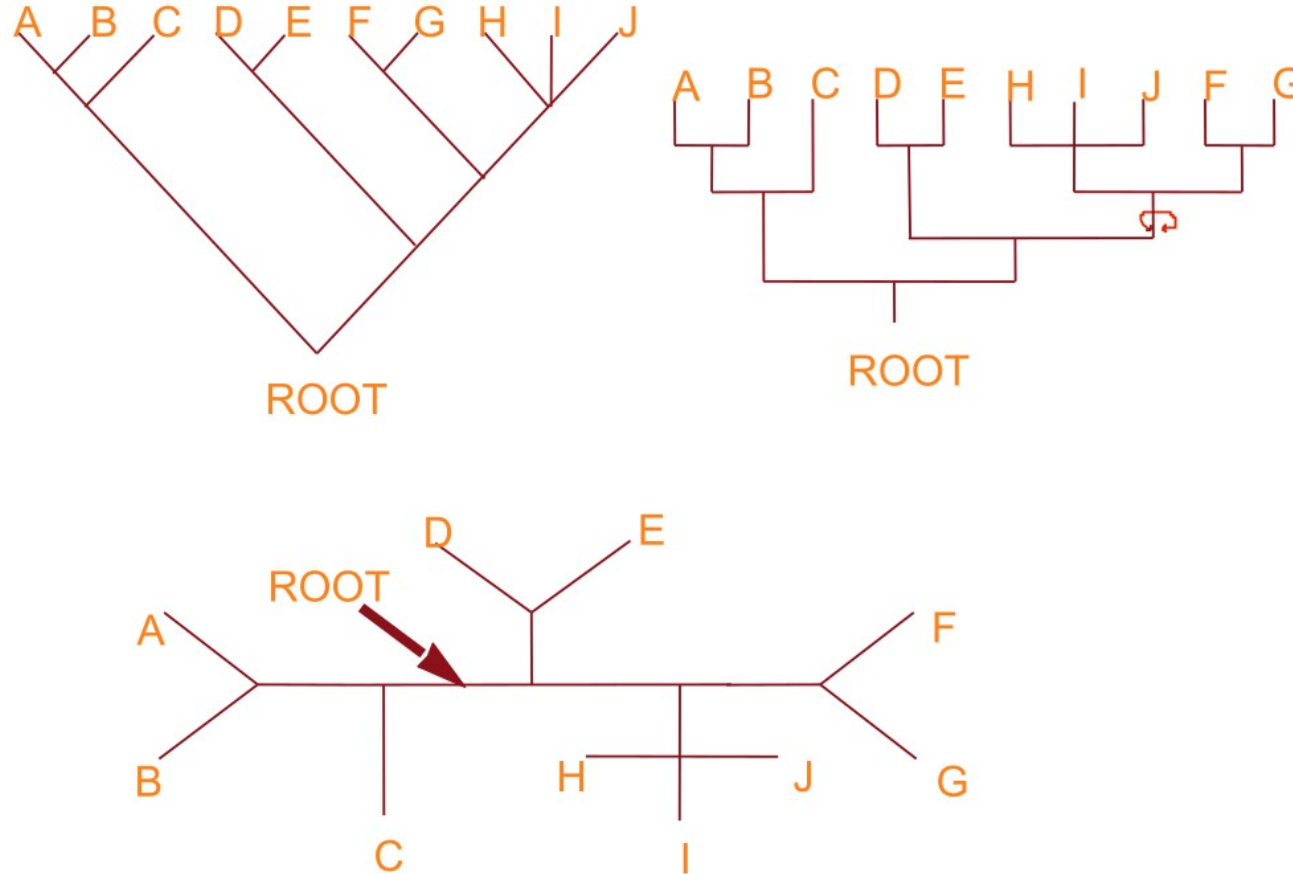


A CLADOGRAM

Example of a complex phylogenetic tree



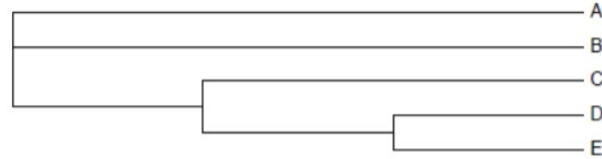
- ▶ **Cladogram** – a tree diagram which depicts a hypothesised evolutionary history
- ▶ **Phylogram** – a tree which indicates by branch length the degree of change believed to have occurred along each lineage
- ▶ **Chronogram** – a tree in which branch lengths are directly in proportion to time



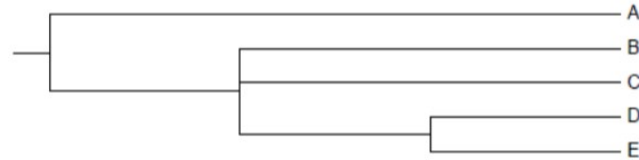
Rooting a tree using an outgroup. Outgroup is a related organism that diverged earlier

The graphical tree code – Newick format

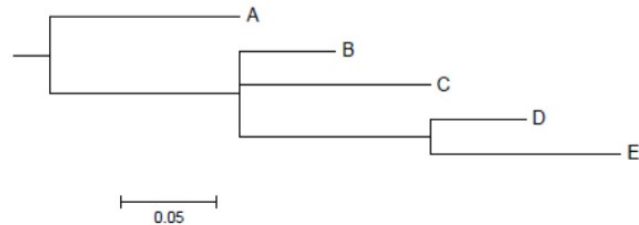
`(A,B,(C,(D,E)));`



`(A,(B,C,(D,E)));`

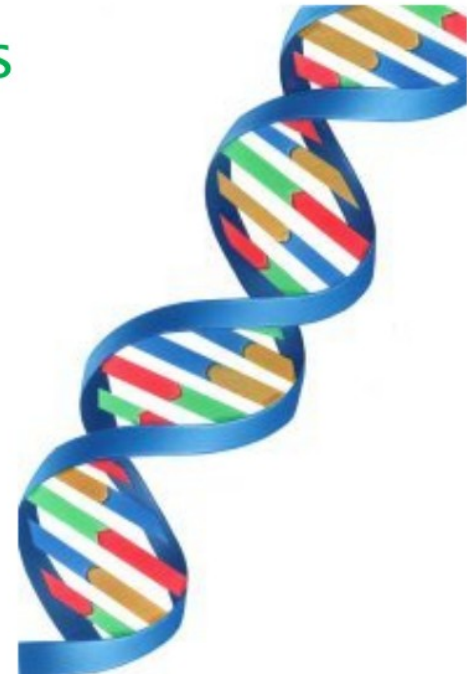


`(A:0.1,(B:0.05,C:0.1,(D:0.05,E:0.1):0.1):0.1);`



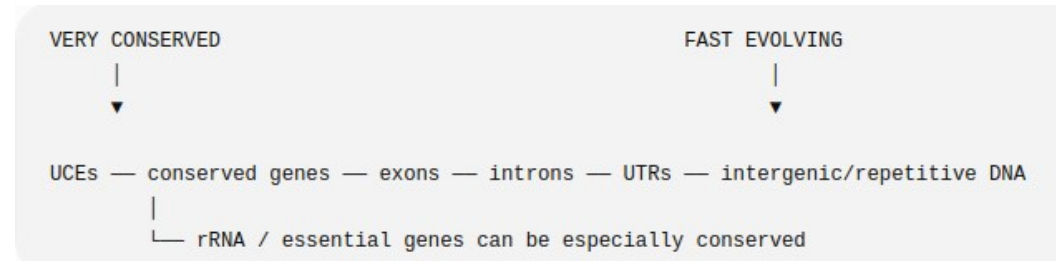
- ▶ Carbohydrates — No
- ▶ Lipids — No
- ▶ Secondary metabolites — No
- ▶ Proteins (amino acids) — Yes
- ▶ Nucleic Acids (DNA and RNA) — Yes

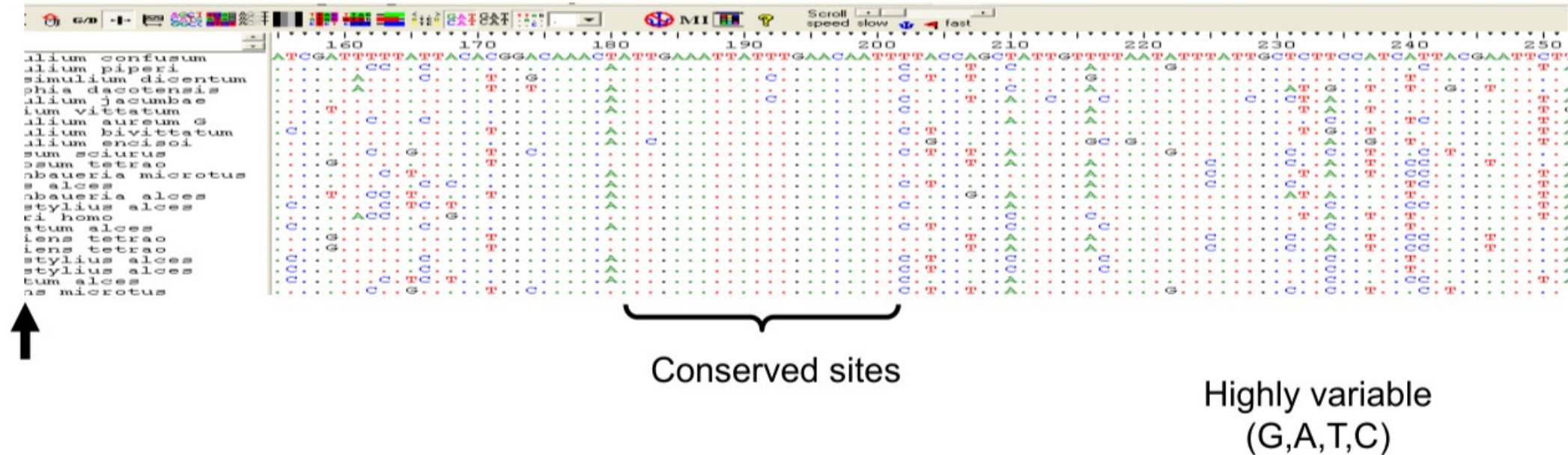
Proteins and DNA are sometimes described as “informational macromolecules”



- ▶ **Choice of sequence** – appropriate for question (fast or slow evolving – close or distant relationships).
- ▶ Many sequences are a **mosaic** of different rates
 - **rRNA** different structural regions evolve at different rates
 - **Proteins** – synonymous (silent) rate (codon position 3) is often faster than nonsynonymous (positions 1 & 2 – changes aa) rate of change
 - Transitions occur more readily than transversions

DNA gradient
mutation rate:



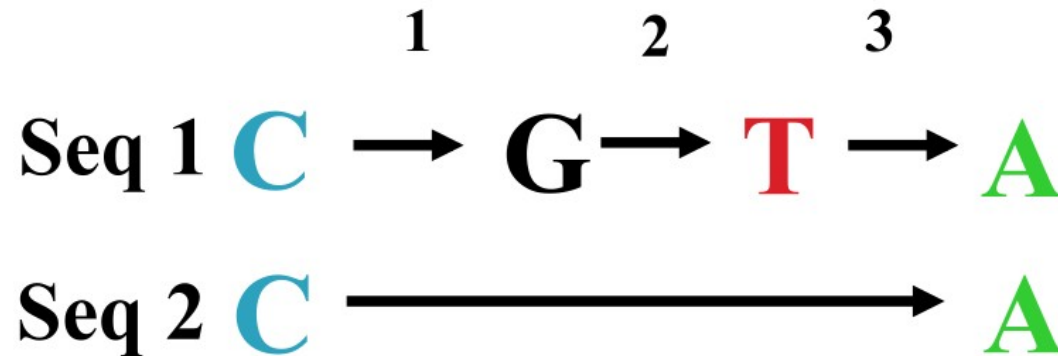


Ancest GGCG**C**G

Seq 1 AGCG**A**G

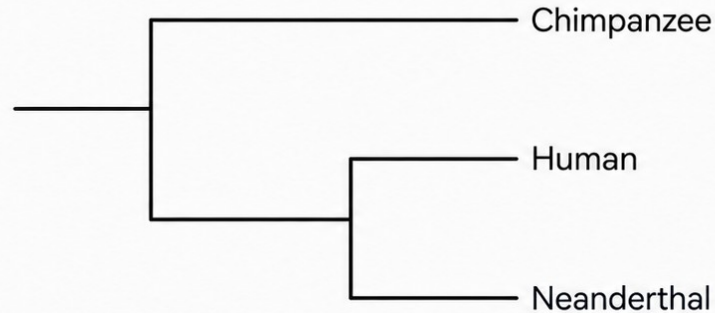
Seq 2 GCGG**A**C

Number of changes



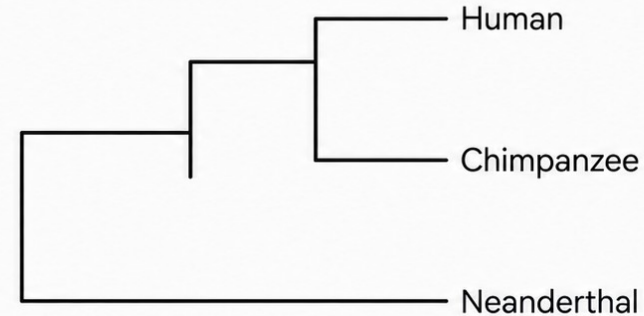
Models of evolution attempt to infer the missing information through correcting for “multiple hits”

Species tree
(actual evolutionary history)



In the species tree, humans and Neanderthals are sister species, and chimpanzees are the outgroup.

Gene tree (due to incomplete lineage sorting)
(one possible gene)



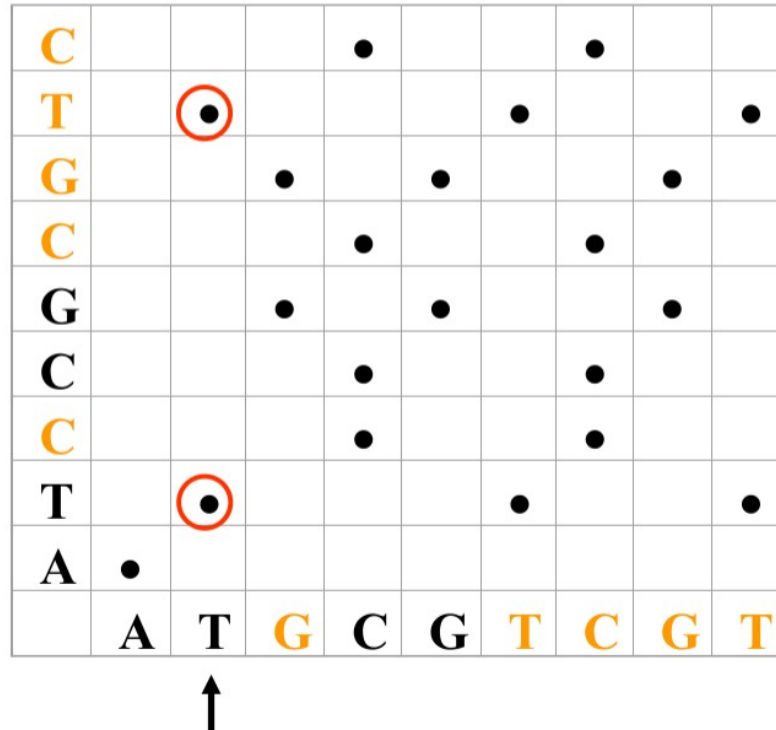
Due to incomplete lineage sorting, a particular gene can show a different branching order, with humans and chimpanzees appearing as the closest relatives.

Gene phylogenetic trees may differ from species phylogenetic trees due to ILS

Alignment of two nucleotide sequences

Two sequences:

ATCCGCGTC
ATGCGTCGT



How do we determine
their relative arrangement,
sliding left/right, gaps...

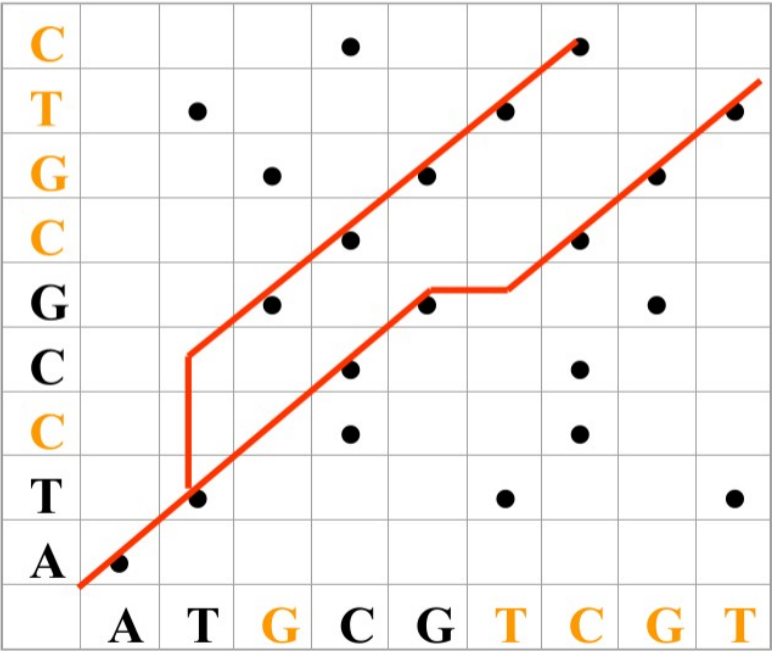
Different alignment scenarios

2

```
ATCCGCGTC
||  |||||
AT--GCGTCGT
```

1

```
ATCCG-CGTC
||  ||  |||
ATGCGTCGT
```



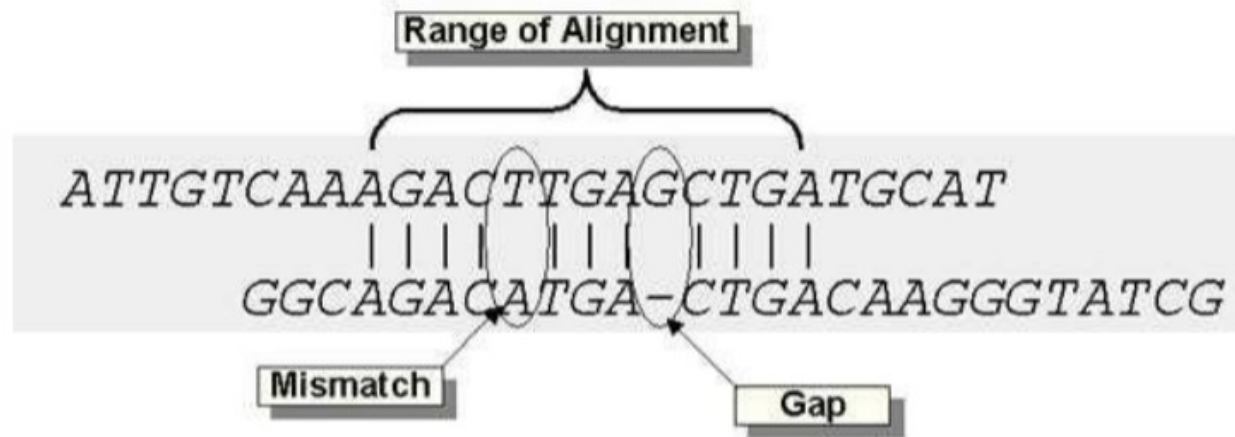
Which alignment
is correct?

Penalty scores

Alignment quality scores computation

Calculation of an alignment score.

The score describes the overall quality of the alignment.



$$S = \sum (\text{identities, mismatches}) - \sum (\text{gap penalties})$$

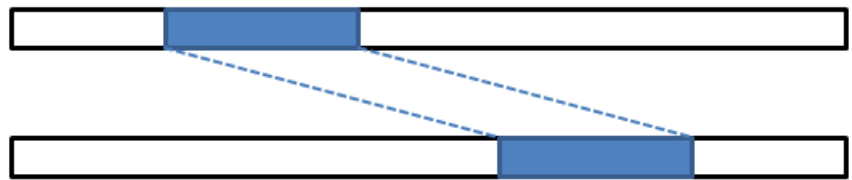
$$\text{Score} = \text{Max}(S)$$

Similarity (homology) is a very strong assumption, which might be totally wrong!

Global vs. local alignment



Global Alignment



Local Alignment

L G P S S K Q T G K G S - S R I W D N
| | | | | | |
L N - I T K S A G K G A I M R L G D A

Global alignment

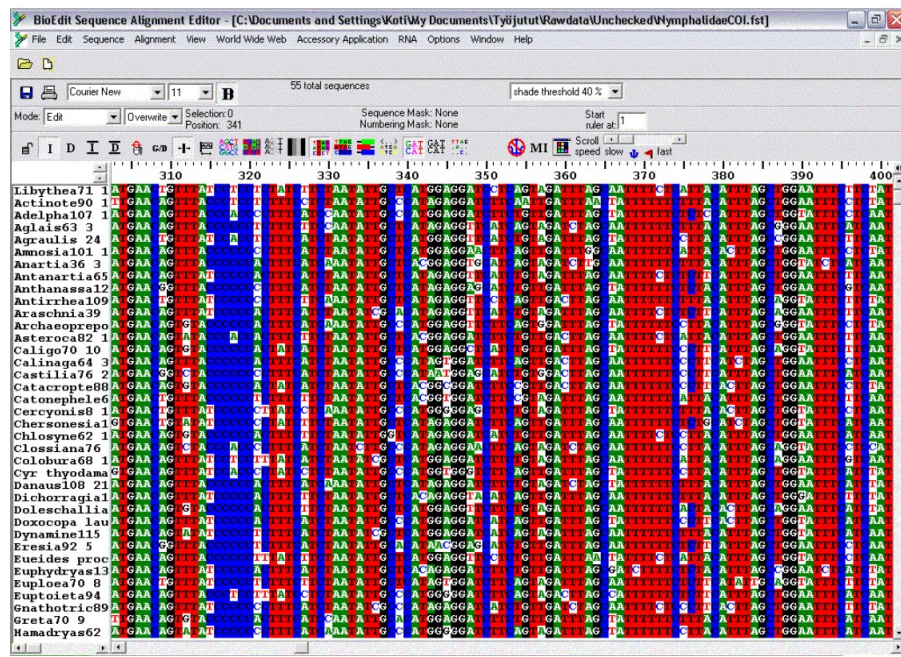
- - - - - T G K G - - - - -
 | | |
- - - - - A G K G - - - - -

Local alignment

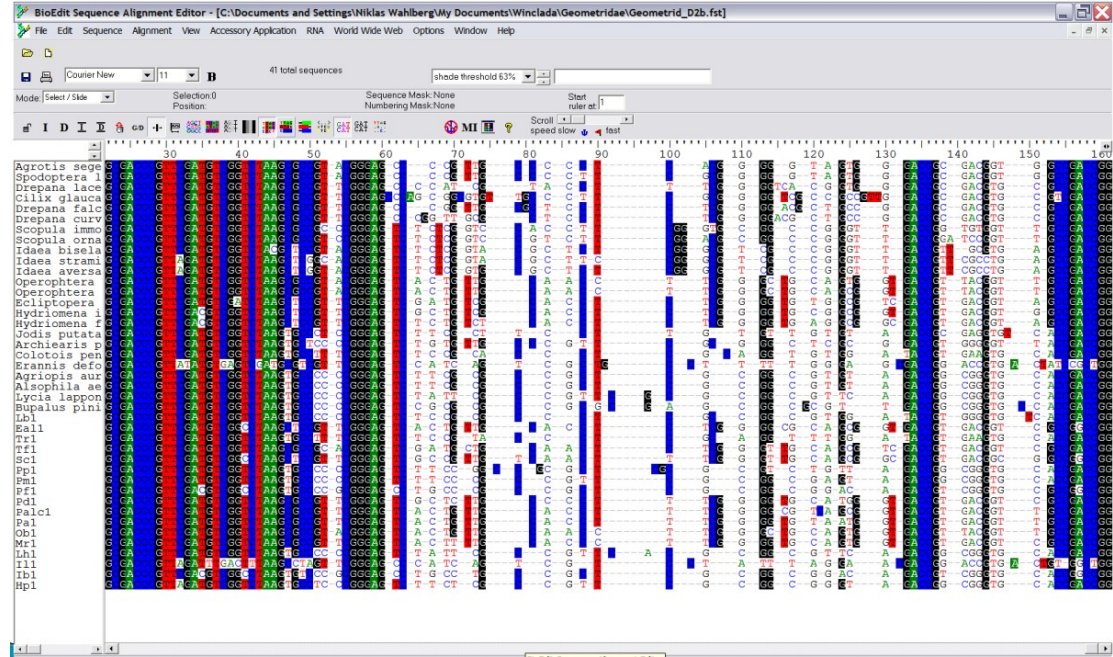
Local: Smith-Waterman

Global: Needleman-Wunsch

Multiple sequence alignment (global)



Easy!



Difficult!

- **Heuristic (approximate) methods**

- **Progressive alignment**

- E.g. Clustal

- **Compute all pairwise alignments**
 - **Align most similar sequences**
 - **Gradually add more sequences**
 - **Fast but prone to propagation of error**

- **Consistency-based scoring**

- Used in e.g. T-Coffee

- **Pairwise alignments are checked against a library of other sequence alignments**
 - **Reduce risk of early mistakes**

- **Iterative refinement methods**

- Used in Mafft

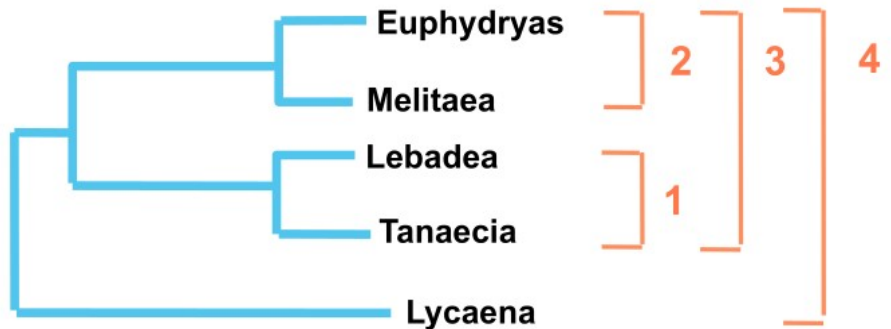
- **Start with initial alignment, change (e.g. delete sequences) and improve**

Clustal multiple sequence alignment

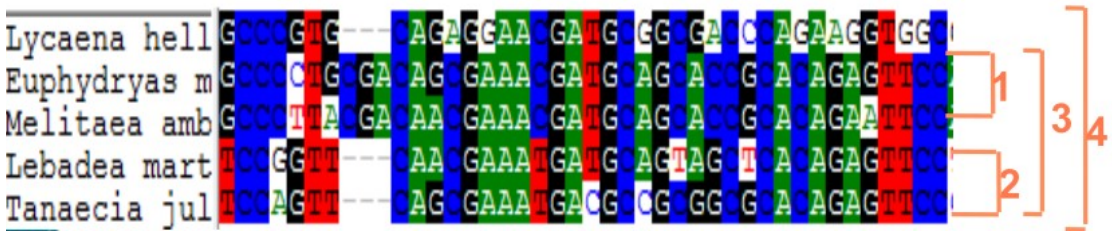


Euphydryas	1	-			
Melitaea	2	.17	-		
Lebadea	3	.59	.60	-	
Tanaecia	4	.59	.59	.13	-
Lycaena	5	.77	.77	.75	.75 -

Quick pairwise alignment:
calculate distance matrix



Neighbour-joining tree
(guide tree)



Progressive alignment
following guide tree

MSA online via Clustal



Address <http://www.ebi.ac.uk/clustalw/>

EMBL-EBI
European Bioinformatics Institute

Get Nucleotide sequences for Go ? Site search Go ?

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Submissions

SEQUENCE ANALYSIS

• Help Index

• General Help

• Formats

• Gaps

• Matrix

• References

• ClustalW Help

ClustalW

Submission Form

Clustal W is a general purpose multiple sequence alignment program for DNA or proteins. It produces biologically meaningful multiple sequence alignments of divergent sequences. It calculates the best match for the selected sequences, and lines them up so that the identities, similarities and differences can be seen. Evolutionary relationships can be seen via viewing Cladograms or Phylograms.

[Download Software](#)

ALIGNMENT TITLE	ALIGNMENT	OUTPUT FORMAT		OUTPUT ORDER	COLOR ALIGNMENT
NONE-	full	aln w/numbers		aligned	no
KTUP (WORD SIZE)	WINDOW LENGTH	SCORE TYPE		TOPDIAG	PAIRGAP
def	def	percent		def	def
MATRIX	GAP OPEN	END GAPS	GAP EXTENSION	GAP DISTANCES	CPU MODE
def	def	def	def	def	single

Enter or Paste a set of Sequences in any supported format :

Help

>OTU1
TTATCCGCGTCGATTTCGATGCGTTAGATGCTGTGCCAAAACATTTT
TT
>OTU2
TTTTTATTCGCGTCGATACGAAGCGTTACCTGCTGAGCGAAATGA
>OTU3
TTTATCCCGTCGATTTCGAAGCGTTACATCCTGTGCCAAAACA

Upload a file:

Browse...

Run

Reset

Fasta filename.fas

```
>falciiparum
NNTAATGGTACTAGGGAATTTTATAGATAATAGAAAATATTTTCATAGAGA
AGTTAACGATTTAGGACCTATTTATGGTTTTCAATGGAGACATTTTCGGTG
>gallinaceu
NNCAACGGACAAGGGAATTTCTAGATAAAAGAAAATATTTTGATAGAGA
AGTTAATGATTTGGGACCCATTTATGGTTTTCAATGGAGACATTTTGGTG
>BRW1
NNAAATGGAACACGTGAATTTTATAGACAACAGGAATTTAACTGACAGAGA
AGTGAATGACTTAGGTCCCATCTATGGATTTTCAGTGGAGACATTTTCGGTG
>GRW1
NNAAATGGAACACGCGAATTTTATAGACAACAGGAATTTAACTGACAGAGA
AGTGAATGACTTAGGTCCCATCTATGGATTTTCAGTGGAGACATTTTCGGTG
>GRW2
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AGTTAATGATTTAGGACCAATTTATGGTTTTTCAGTGGAGACATTTTGGTG
>GRW5
NNAAATGGAACACGCGAATTTTATAGACAACAGGAATTTAACTGACAGAGA
AGTGAATGACTTAGGTCCCATCTATGGATTTCAATGGAGACATTTTCGGTG
>GRW8
NNAAATGGAACACGCGAATTTTATAGACAACAGGAATTTAACTGACAGAGA
AGTGAATGACTTAGGTCCCATCTATGGATTTTCAGTGGAGACATTTTCGGTG
>SYAT1
NN-----TTTTTATAGACAACAGAAAATTTAAACCGACAGAGA
AGTGAATGATTTAGGTCCCATATATGGTTTTTCAATGGAGACACTTTCGGTG
>SYAT16
NN-----TTTTTATAGACAACAGGAATTTAACTGACAGAGA
AGTGAATGATTTAGGTCCCATATATGGTTTTTCAATGGAGACACTTTCGGTG
>SYAT3
NN-----TTTTTGGATAAAAGAAAATCTATGATAGAGA
GGTAAATGATTTGGGACCAATATATGGTTTTTTCAGTGGAGACATTTTCGGAG
```

Phylip filename.phy

```
17 238
falciiparum      NNTAATGGTA CTAGGGAATT TTTAGATAAT AGAAAATTAT TTCATAGAGA
gallinaceu       NNCAACGGAA CAAGGGAATT CTTAGATAAA AGAAAATTAT TTGATAGAGA
BRW1             NNAAATGGAA CACGTGAATT TTTAGACAAC AGGAATTTAA CTGACAGAGA
GRW1             NNAAATGGAA CACGCGAATT TTTAGACAAC AGGAATTTAA CTGACAGAGA
GRW2             NNTAATGGCA CGAGAGAATT TTTAGATAGT CGAAAATCTAT TTGATAGAGA
GRW5             NNAAATGGAA CACGCGAATT TTTAGACAAC AGGAATTTAA CTGACAGAGA
GRW8             NNAAATGGAA CACGCGAATT TTTAGACAAC AGGAATTTAA CTGACAGAGA
SYAT1            NN----- TTTAGACAAC AGAAATTTAA CCGACAGAGA
SYAT16           NN----- TTTAGACAAC AGGAATTTAA CTGACAGAGA
SYAT3            NN----- TTTGGATAAA AGAAAATCTGT ATGATAGAGA
SYAT7            NN----- TTTAGACAAC AGGAATTTAA CTGACAGAGA
WW1             NNTAACGGAA CTAGAGAATT TTTAGATAAA AGGAAATTTGT ATGATAGAGA
vivax            NNAAATGGAA CGAGGGAGTT CCTCGACAAC AGGAAATTTAT TCCACAGAGA
chabaudi         NNTAATGGGA CAAGAGAGTT TTTAGATAAT AGGAAATTTAT TTCATAGAGA
berghei          NNTAATGGAA CAAGGGAATT TTTAGATAAT AGAAAATTAT TTCATAGAGA
SGS1             NN----- TTTAGATAAT AGAAAATTAT TCGATAGAGA
PARUS1           NNTAATGGAA CCAGAGAATT TTTAGACAAC AGAAATTTAT TTGATAGAGA

AGTTAACGAT TTAGGACCTA TTTATGGTTT TCAATGGAGA CATTTTCGGTG
AGTTAATGAT TTGGGACCCA TTTATGGTTT TCAATGGAGA CATTTTGGTG
AGTGAATGAC TTAGGTCCCA TCTATGGATT TCAGTGGAGA CATTTTCGGTG
AGTGAATGAC TTAGGTCCCA TCTATGGATT TCAGTGGAGA CATTTTCGGTG
AGTTAATGAT TTAGGACCAA TTTATGGTTT TCAGTGGAGA CATTTTGGTG
AGTGAATGAC TTAGGTCCCA TCTATGGATT TCAATGGAGA CATTTTCGGTG
AGTGAATGAC TTAGGTCCCA TCTATGGATT TCAGTGGAGA CATTTTCGGTG
AGTGAATGAT TTAGGTCCCA TATATGGTTT TCAATGGAGA CACTTTCGGTG
AGTGAATGAT TTAGGTCCCA TATATGGTTT TCAATGGAGA CACTTTCGGTG
GGTAAATGAT TTGGGACCAA TATATGGTTT TCAGTGGAGA CATTTTCGGAG
AGTGAATGAT TTAGGTCCCA TATATGGTTT TCAATGGAGA CACTTTCGGTG
GGTAAATGAT TTGGGACCAA TATATGGTTT TCAGTGGAGA CATTTTCGGTG
AGTGAATGAC TTGGGGCCCA TTTATGGCTT CCAGTGGAGA CACTTTCGGTG
AGTTAATGAC TTAGGTCCAA TTTACGGATT TCAATGGAGG CACTTTGGTG
AGTTAATGAT CTCGGTCCAA TTTATGGATT TCAATGGAGG CATTTTGGTG
AGTTAATGAT TTGGGACCCA TTTATGGTTT TCAATGGAGA CATTTTCGGTG
AGTGAATGAT TTAGGTCCCA TTTATGGATT TCAGTGGAGA CATTTTCGGTG
```

OTU1 ATCCGCGTCGATTCGATGCGTTAGATGCTGTGCCAAAACA
 OTU2 ATTC^{CGTCGAT}ACGA^{AGCGTTAC}CTGCTG^{AGC}GAAATGA
 OTU3 ATCC^{CGTCGATTCGA}AGCGTTAC^{AT}CCTGTGCCAAAACA

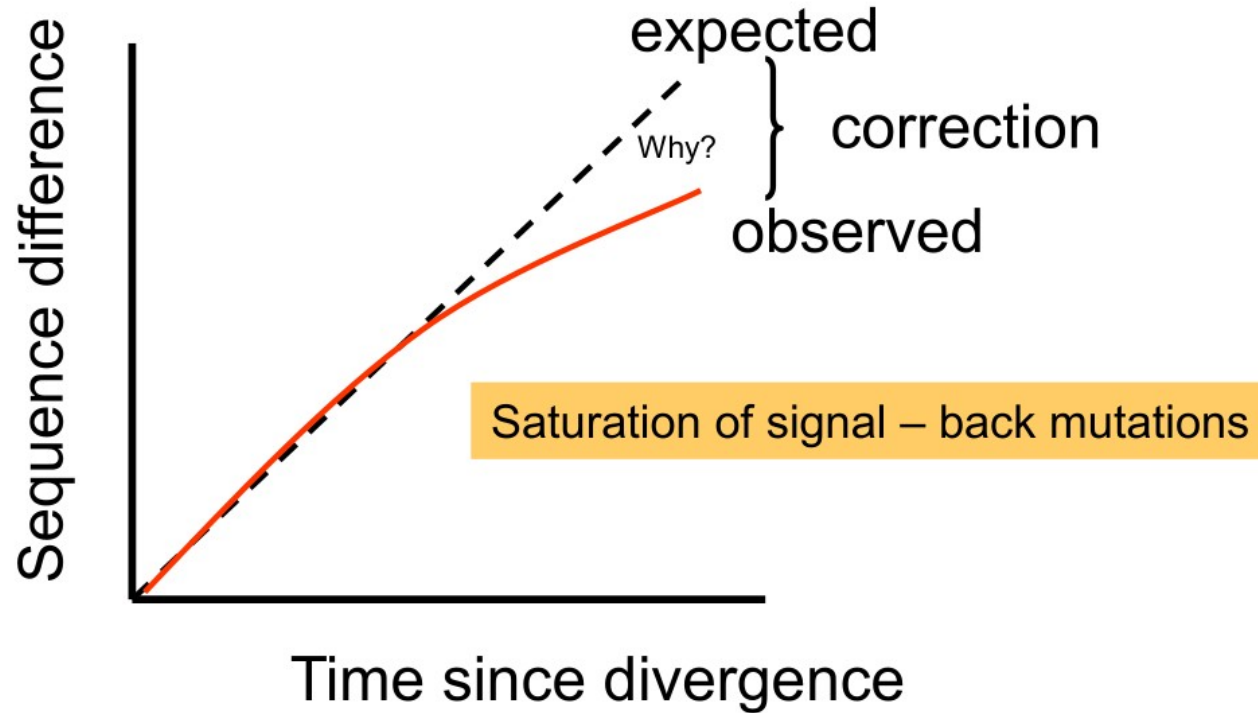
OTU = operational taxonomic unit

1. Number of differences

	OTU1	OTU2	OTU3
OTU1		10	4
OTU2	25%		8
OTU3	10%	20%	

But what does the differences mean?

Corrected genetic distance



True (d_T) vs. Observed distance (d_O)

- $d_T \geq d_O$
- $d_O = N/L$ where N = no. dif. and L = no. sites
- $d_T = S/L$ where S = no. of actual substitutions
- S can not be observed but estimated using evolutionary models

Model of nucleotide sequence evolution

$$P_t = \begin{pmatrix} \begin{matrix} & A & C & G & T \\ \begin{matrix} A \\ C \\ G \\ T \end{matrix} & \begin{matrix} p_{AA} & p_{AC} & p_{AG} & p_{AT} \\ p_{CA} & p_{CC} & p_{CG} & p_{CT} \\ p_{GA} & p_{GC} & p_{GG} & p_{GT} \\ p_{TA} & p_{TC} & p_{TG} & p_{TT} \end{matrix} \end{matrix} \end{pmatrix}$$

Substitution
probability matrix

$$\mathbf{f} = [f_A \ f_C \ f_G \ f_T]$$

Base composition
vector

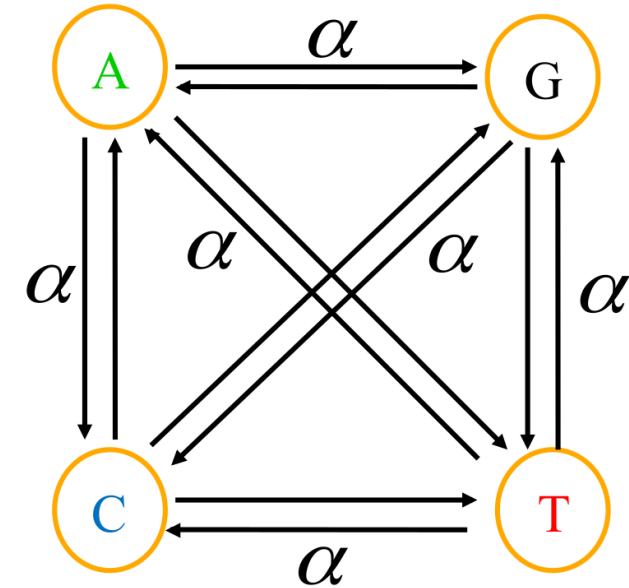
Model of nucleotide sequence evolution



Table 1.1 Substitution-rate matrices for commonly used Markov models of nucleotide substitution

	From	To			
		T	C	A	G
JC69 (Jukes and Cantor 1969)	T	.	λ	λ	λ
	C	λ	.	λ	λ
	A	λ	λ	.	λ
	G	λ	λ	λ	.
K80 (Kimura 1980)	T	.	α	β	β
	C	α	.	β	β
	A	β	β	.	α
	G	β	β	α	.
F81 (Felsenstein 1981)	T	.	π_C	π_A	π_G
	C	π_T	.	π_A	π_G
	A	π_T	π_C	.	π_G
	G	π_T	π_C	π_A	.
HKY85 (Hasegawa <i>et al.</i> 1984, 1985)	T	.	$\alpha\pi_C$	$\beta\pi_A$	$\beta\pi_G$
	C	$\alpha\pi_T$	.	$\beta\pi_A$	$\beta\pi_G$
	A	$\beta\pi_T$	$\beta\pi_C$	.	$\alpha\pi_G$
	G	$\beta\pi_T$	$\beta\pi_C$	$\alpha\pi_A$	.
F84 (Felsenstein, DNAML program since 1984)	T	.	$(1 + \kappa/\pi_Y)\beta\pi_C$	$\beta\pi_A$	$\beta\pi_G$
	C	$(1 + \kappa/\pi_Y)\beta\pi_T$	.	$\beta\pi_A$	$\beta\pi_G$
	A	$\beta\pi_T$	$\beta\pi_C$	.	$(1 + \kappa/\pi_R)\beta\pi_G$
	G	$\beta\pi_T$	$\beta\pi_C$	$(1 + \kappa/\pi_R)\beta\pi_A$	.
TN93 (Tamura and Nei 1993)	T	.	$\alpha_1\pi_C$	$\beta\pi_A$	$\beta\pi_G$
	C	$\alpha_1\pi_T$	.	$\beta\pi_A$	$\beta\pi_G$
	A	$\beta\pi_T$	$\beta\pi_C$	.	$\alpha_2\pi_G$
	G	$\beta\pi_T$	$\beta\pi_C$	$\alpha_2\pi_A$	.
GTR (REV) (Tavaré 1986; Yang 1994 <i>b</i> ; Zharkikh 1994)	T	.	$a\pi_C$	$b\pi_A$	$c\pi_G$
	C	$a\pi_T$	.	$d\pi_A$	$e\pi_G$
	A	$b\pi_T$	$d\pi_C$	.	$f\pi_G$
	G	$c\pi_T$	$e\pi_C$	$f\pi_A$	.
UNREST (Yang 1994 <i>b</i>)	T	.	q_{TC}	q_{TA}	q_{TG}
	C	q_{CT}	.	q_{CA}	q_{CG}
	A	q_{AT}	q_{AC}	.	q_{AG}
	G	q_{GT}	q_{GC}	q_{GA}	.

The diagonals of the matrix are determined by the requirement that each row sums to 0. The equilibrium distribution is $\pi = (1/4, 1/4, 1/4, 1/4)$ under JC69 and K80, and $\pi = (\pi_T, \pi_C, \pi_A, \pi_G)$ under F81, F84, HKY85, TN93, and GTR. Under the general unrestricted (UNREST) model, it is given by the equations $\pi Q = 0$ under the constraint $\sum_i \pi_i = 1$.

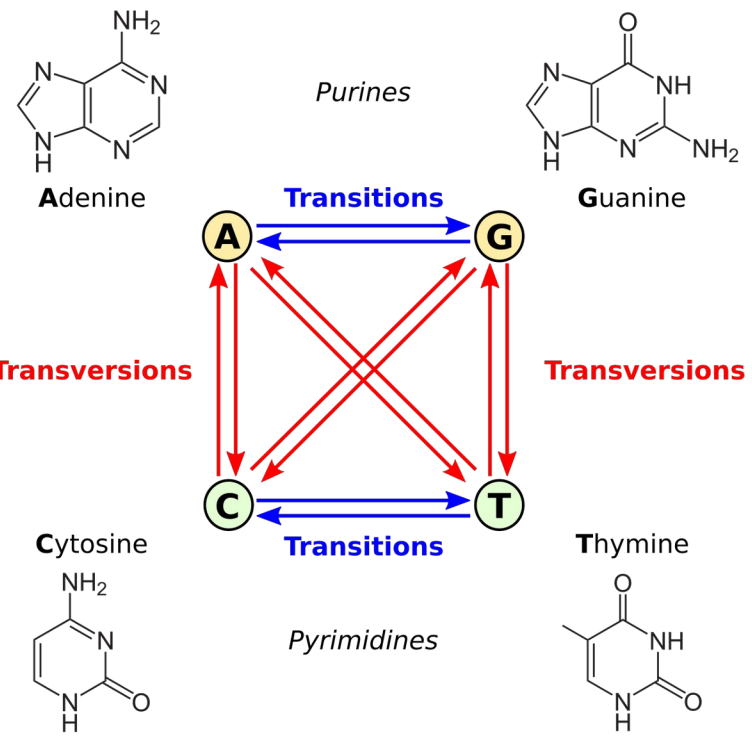


$$P_t = \left\{ \begin{array}{c|c|c|c} \bullet & \alpha & \alpha & \alpha \\ \alpha & \bullet & \alpha & \alpha \\ \alpha & \alpha & \bullet & \alpha \\ \alpha & \alpha & \alpha & \bullet \end{array} \right\}$$

$$f = [\frac{1}{4} \frac{1}{4} \frac{1}{4} \frac{1}{4}]$$

All changes count equal!

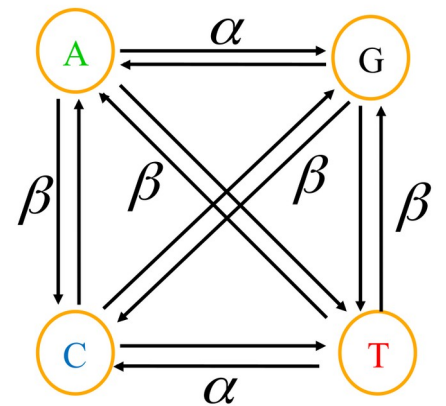
Kimura (K80) model



$$P_t = \left\{ \begin{array}{|c|c|c|c|} \hline \bullet & \beta & \alpha & \beta \\ \hline \beta & \bullet & \beta & \alpha \\ \hline \alpha & \beta & \bullet & \beta \\ \hline \beta & \alpha & \beta & \bullet \\ \hline \end{array} \right\}$$

$$f = [\frac{1}{4} \frac{1}{4} \frac{1}{4} \frac{1}{4}]$$

α = transitions
 β = transversions



	OTU1- OTU2	OTU1- OTU3	OTU2- OTU3
p- distance	0.25	0.10	0.20
JC	0.304	0.107	0.233
K2P	0.310	0.108	0.235

$$d_{JC} = -3/4 \times \ln(1 - 4/3 \times p)$$

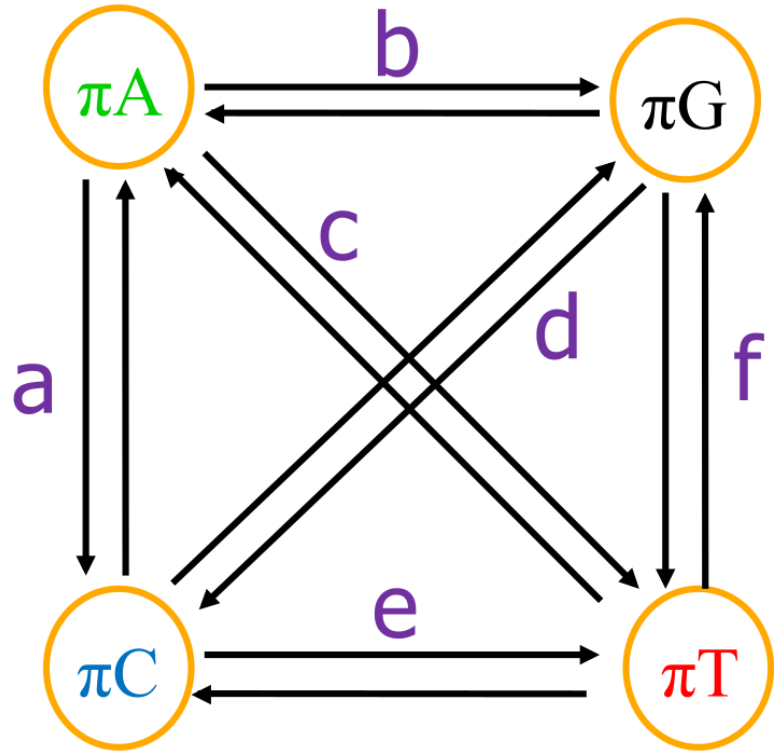
Not correcting for saturation

All mutations = same

Transitions $\neq$ transversions

Corrected genetic distance is greater than the Hamming distance!

General time reversible (GTR) model

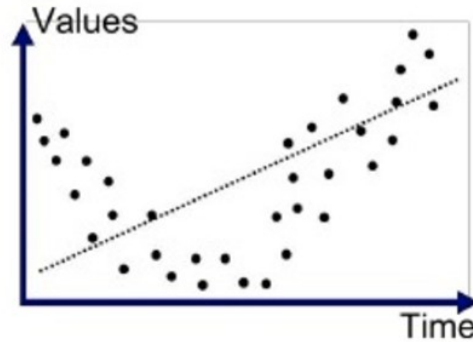


	A	C	G	T
A	—	$\pi_C a$	$\pi_G b$	$\pi_T c$
C	$\pi_A a$	—	$\pi_G d$	$\pi_T e$
G	$\pi_A b$	$\pi_C d$	—	$\pi_T f$
T	$\pi_A c$	$\pi_C e$	$\pi_G f$	—

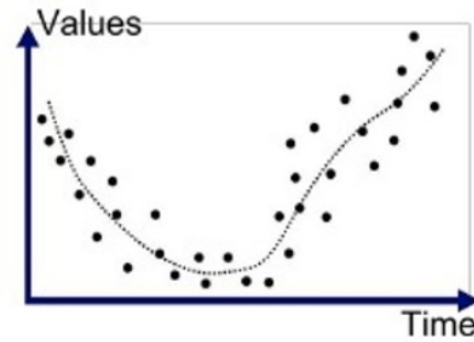
GTRmodel:

- 1) Base frequencies are allowed to vary: π_A , π_C , π_G , π_T
- 2) There are six substitution types: a , b , c , d , e , f

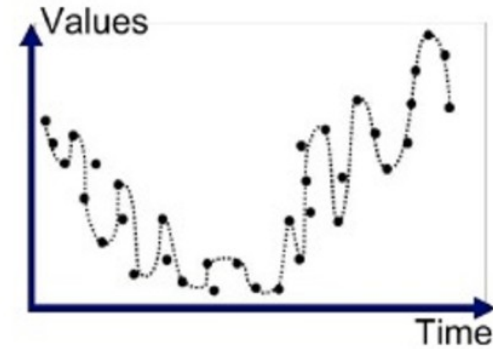
All other models are special cases of GTR



Underfitted



Good Fit/Robust



Overfitted

- Akaike Information Criterion (AIC):

$$AIC = 2k - 2\ln(L)$$

- Random Forest: each tree overfitted, but ensemble of trees performs very well

Likelihood Ratio Test (LRT)



$$LR = 2 * (\ln L_1 - \ln L_0)$$

Alternative hypothesis

More parameter-rich

Null hypothesis

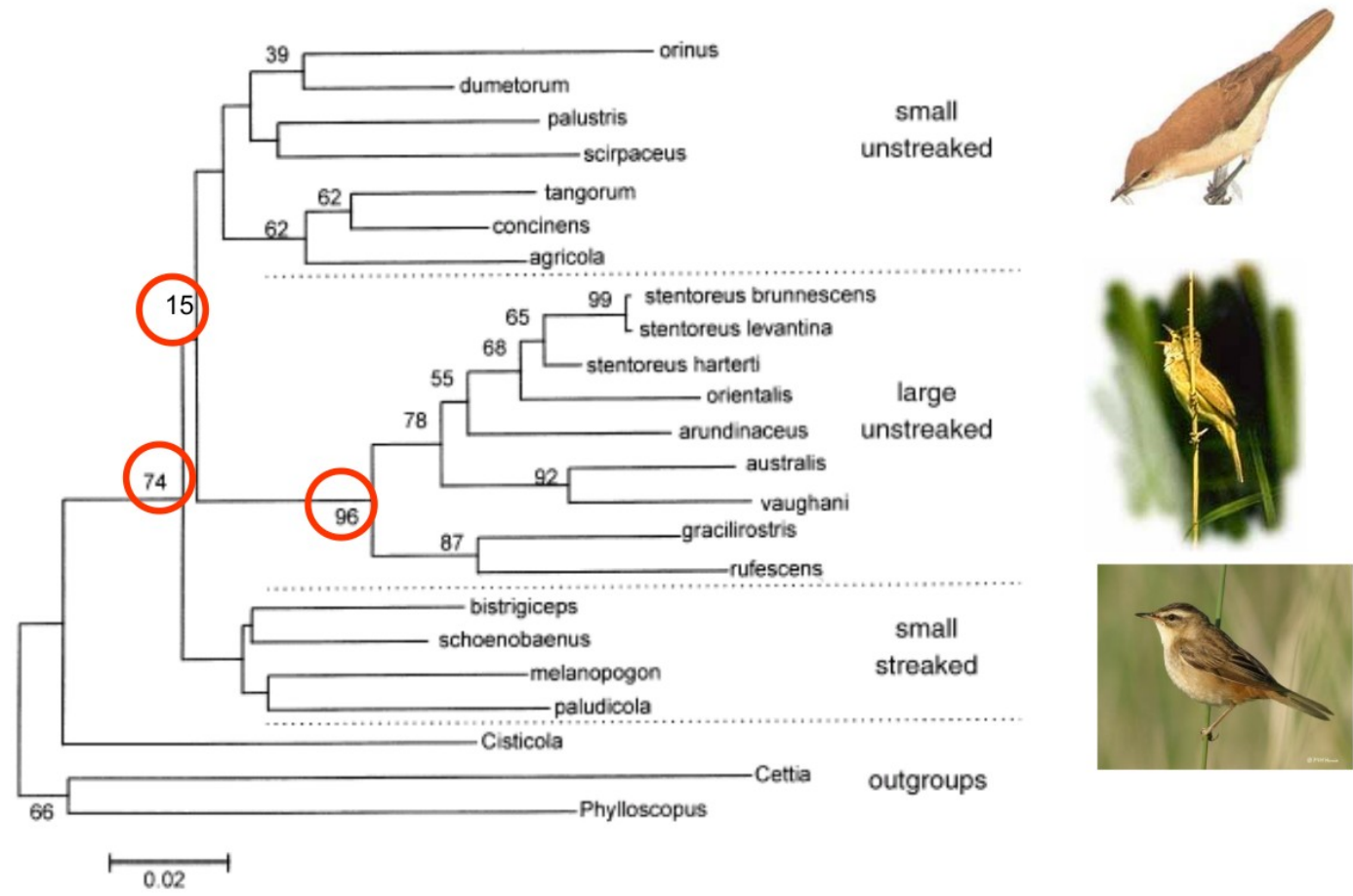
Less parameter-rich

- LRT statistic approximately follows a chi-square distribution
- Degrees of freedom equal to the number of extra parameters in the more complex model

- A measure of the **relative** quality of statistical models for a given dataset (Wikipedia definition)
 - It deals with the trade-off between the goodness of fit and the complexity of the model
- $AIC(M) = -2 * \text{Log}(\text{Likelihood}(M)) + 2 * K(M)$
 - $K(M)$ is the number of estimable parameters of model M
- Given a dataset, models can be ranked according to their AIC
- The model with the lowest AIC is selected
- AIC_c – correction for finite sample size – usually used

- BIC takes into account also sample size n
- $\text{BIC}(\mathbf{M}) = -2 * \text{Log}(\text{Likelihood}(\mathbf{M})) + K(\mathbf{M}) * \text{Log}(n)$
 - $K(\mathbf{M})$ is the number of estimable parameters of model $\mathbf{M}$ and n is the number of characters
- The model with the lowest BIC is selected

Phylogenetic tree with bootstrapping





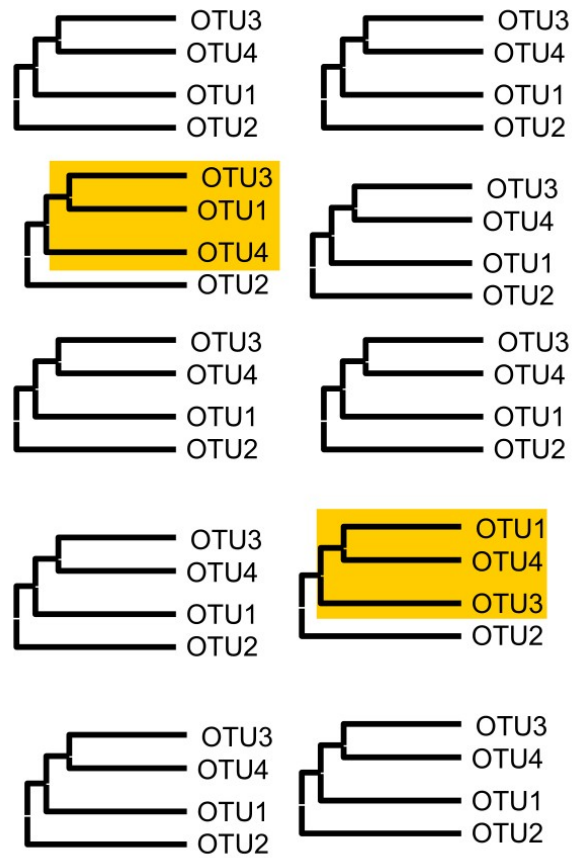
18 17 7 14 4 5 17 21 13 11 5 9 14 20 10 19 10 1 6 1 3



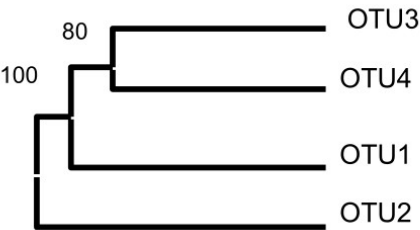
Position 17 twice

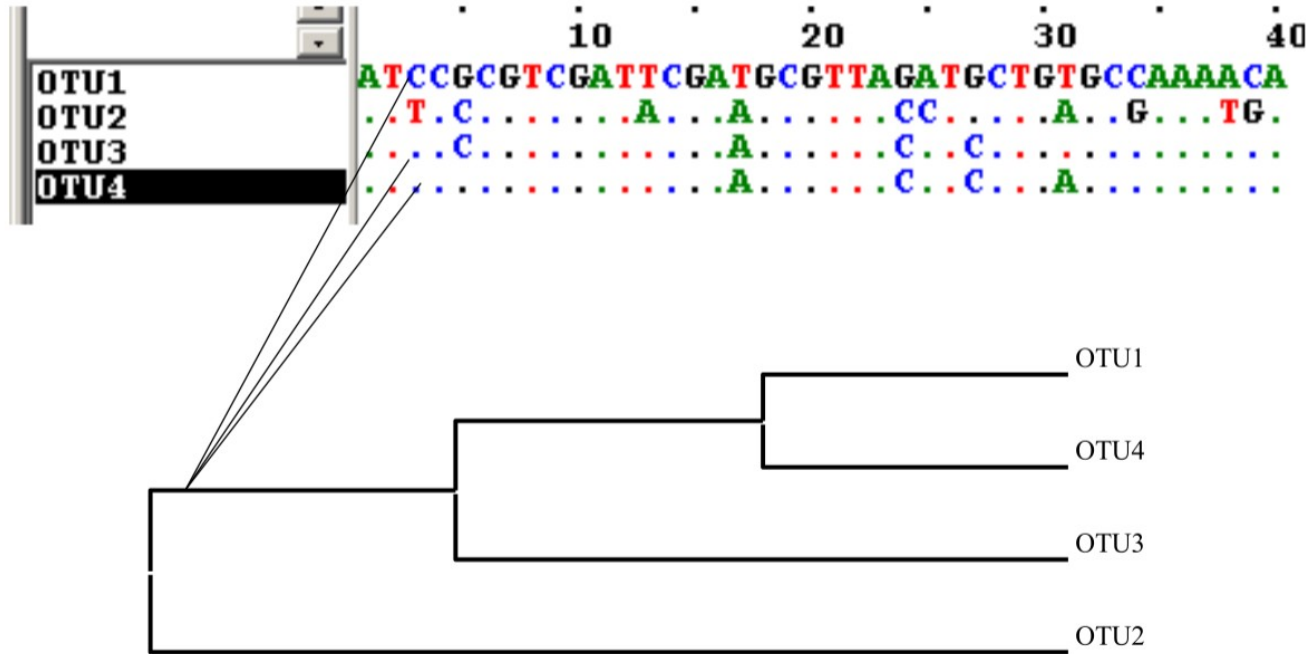
Resamples phylogenetic trees

10 "sampled" Trees



Observed tree
with bootstrap support





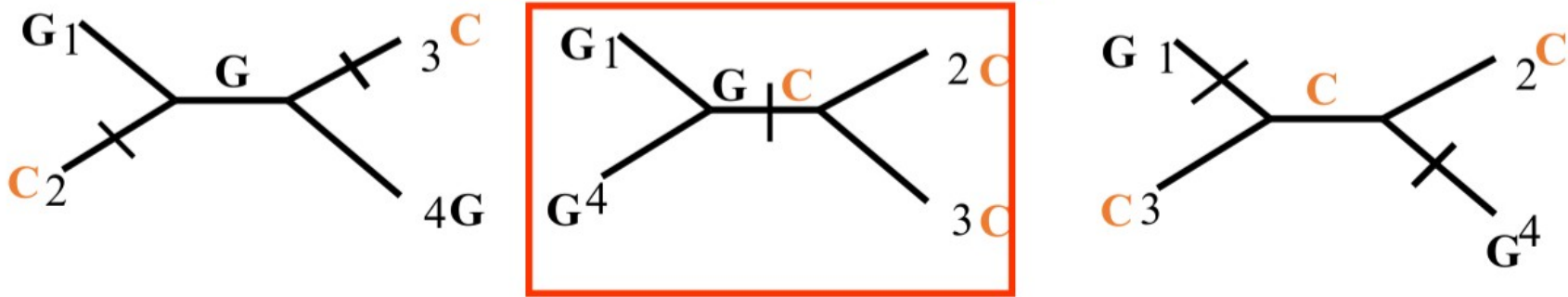
Methods for tree construction

- Neighbor-joining
- Maximum Parsimony
- Maximum likelihood
- Bayesian inference

Chooses the tree that requires the fewest evolutionary changes



3 possible unrooted trees



For 5 sequences there are 15 unrooted trees.....

for 20 sequences there are 8×10^{21} unrooted trees...

for 50 sequences there are 3×10^{74} unrooted trees!!!

Particles in universe $\approx 10^{72} - 10^{87}$

$$T_N = \frac{(2N-5)!}{2^{N-3} (N-3)!}$$

... and for each unrooted tree, there are N rooted trees.

How do we go from an multiple alignment to one tree?

1	1
2	1
3	1
4	3
5	15
6	105
7	945
8	10395
9	135135
10	2027025
11	34459425
12	654729075
13	13749310575
14	316234143225
15	7905853580625
16	213458046676875
17	6190283353629370
18	191898783962510625
19	6332659870762850625
20	221643095476699771875 (2×10^{20})
50	3×10^{74}

How can
we find
the most
optimal
tree?

Exhaustive search is very resource demanding and time consuming!

Maximum Likelihood principle



$$L(x_i | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp -\frac{\sum_{i=1}^N (x_i - \mu)^2}{2\sigma^2}$$

$$\frac{\partial L(x_i | \mu, \sigma^2)}{\partial \mu} = 0; \quad \frac{\partial L(x_i | \mu, \sigma^2)}{\partial \sigma^2} = 0$$

$$\mu = \frac{1}{N} \sum_{i=0}^N x_i \text{ -- mean estimator}$$

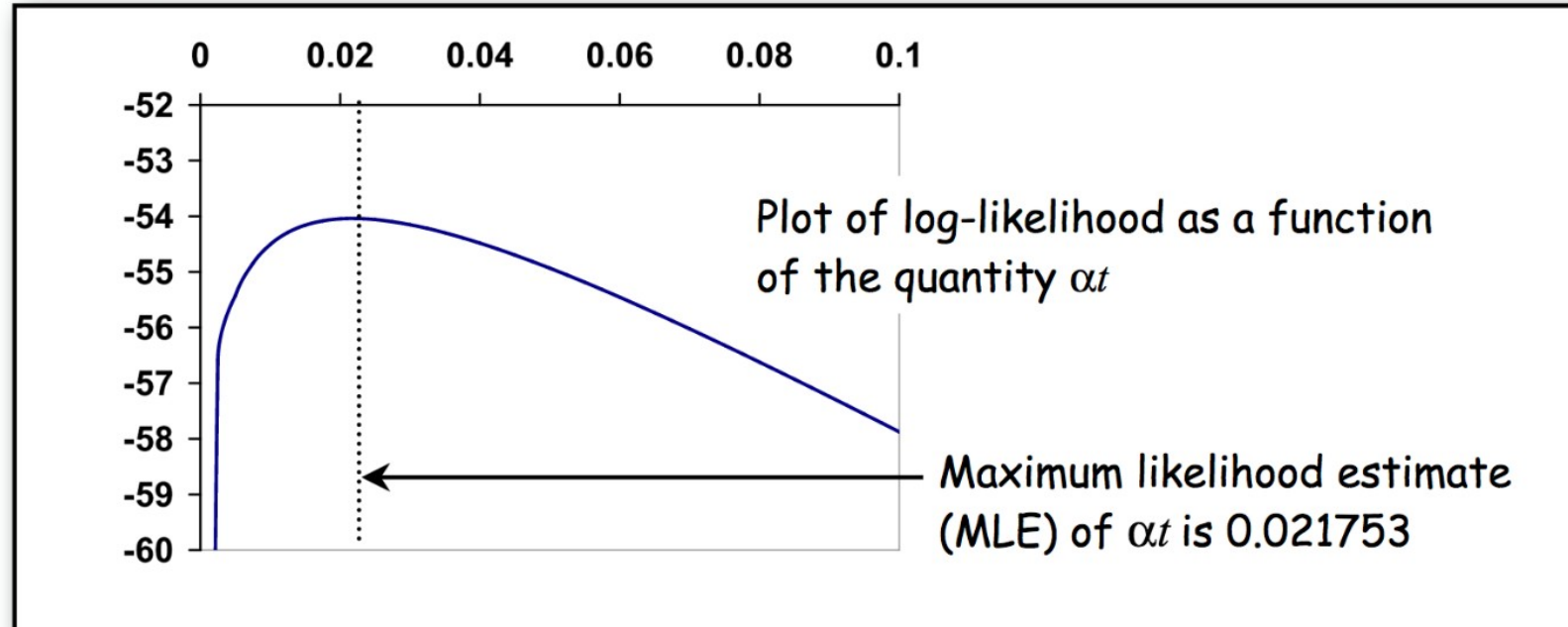
$$\sigma^2 = \frac{1}{N} \sum_{i=0}^N (x_i - \mu)^2 \text{ -- variance estimator}$$

First 32 nucleotides of the $\psi\eta$ -globin gene of gorilla and orangutan:

gorilla **GAAG**TCCTTGAGAAATAAACTGCACACACTGG

orangutan **GGAC**TCCTTGAGAAATAAACTGCACACACTGG

$$L = \left[\left(\frac{1}{4} \right) \left(\frac{1}{4} + \frac{3}{4} e^{-4\alpha t} \right) \right]^{30} \left[\left(\frac{1}{4} \right) \left(\frac{1}{4} - \frac{1}{4} e^{-4\alpha t} \right) \right]^2$$



Here we apply the theory to estimation of the distance between two sequences under the JC69 model (Jukes and Cantor 1969). The single parameter is the distance d . The data are two aligned sequences, each n sites long, with x differences. From equation (1.5), the probability that a site has different nucleotides between two sequences separated by distance d is

$$p = 3p_1 = \frac{3}{4} - \frac{3}{4}e^{-4d/3}. \quad (1.41)$$

Thus, the probability of observing the data, that is, x differences out of n sites, is given by the binomial probability

$$L(d; x) = f(x|d) = Cp^x(1-p)^{n-x} = C \left(\frac{3}{4} - \frac{3}{4}e^{-4d/3} \right)^x \left(\frac{1}{4} + \frac{3}{4}e^{-4d/3} \right)^{n-x}. \quad (1.42)$$

Furthermore, the likelihood L is typically extremely small and awkward to work with. Thus its logarithm $\ell(d) = \log\{L(d)\}$ is commonly used instead. As the logarithm function is monotonic, we achieve the same result; that is, $L(d_1) > L(d_2)$ if and only if $\ell(d_1) > \ell(d_2)$. The *log likelihood function* is thus

$$\ell(d; x) = \log\{L(d; x)\} = x \log \left(\frac{1}{16} - \frac{1}{16}e^{-4d/3} \right) + (n-x) \log \left(\frac{1}{16} + \frac{3}{16}e^{-4d/3} \right). \quad (1.44)$$

To estimate d , we maximize L or equivalently its logarithm ℓ . By setting $d\ell/dd = 0$, we can determine that ℓ is maximized at the MLE

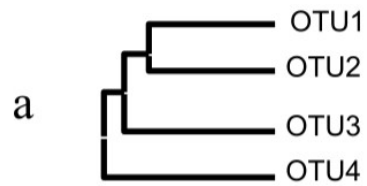
$$\hat{d} = -\frac{3}{4} \log \left(1 - \frac{4}{3} \frac{x}{n} \right). \quad (1.45)$$

This is the distance formula (1.6) under JC69 that we derived earlier.

The necessary condition of likelihood extremum is zero first derivative!

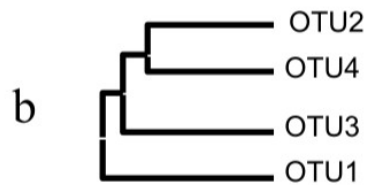
Ziheng Yang, Computational Molecular Evolution 2006

Phylogenetic tree with highest likelihood

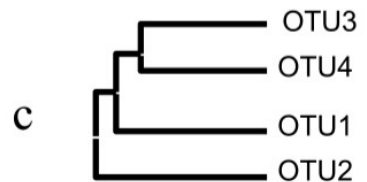


$$-\ln L = 125.17$$

$$= e^{-125.17}$$



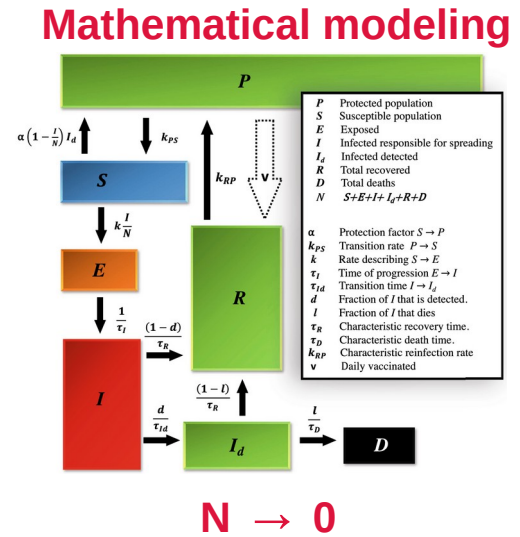
$$-\ln L = 123.25$$



$$-\ln L = 122.75$$

highest likelihood of data!!!

How do we go from an multiple alignment to one tree?



Bayesianism

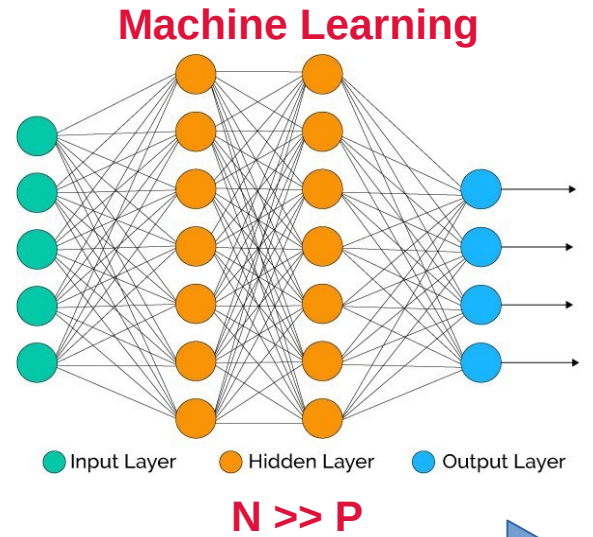


$N \ll P$

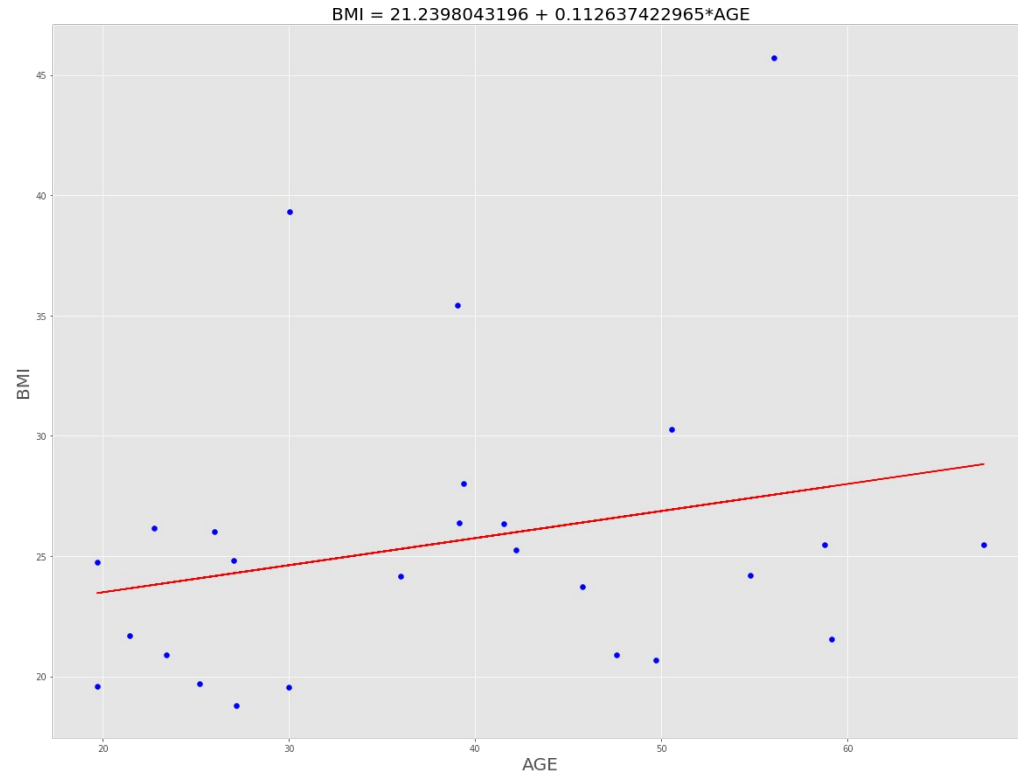
Frequentism



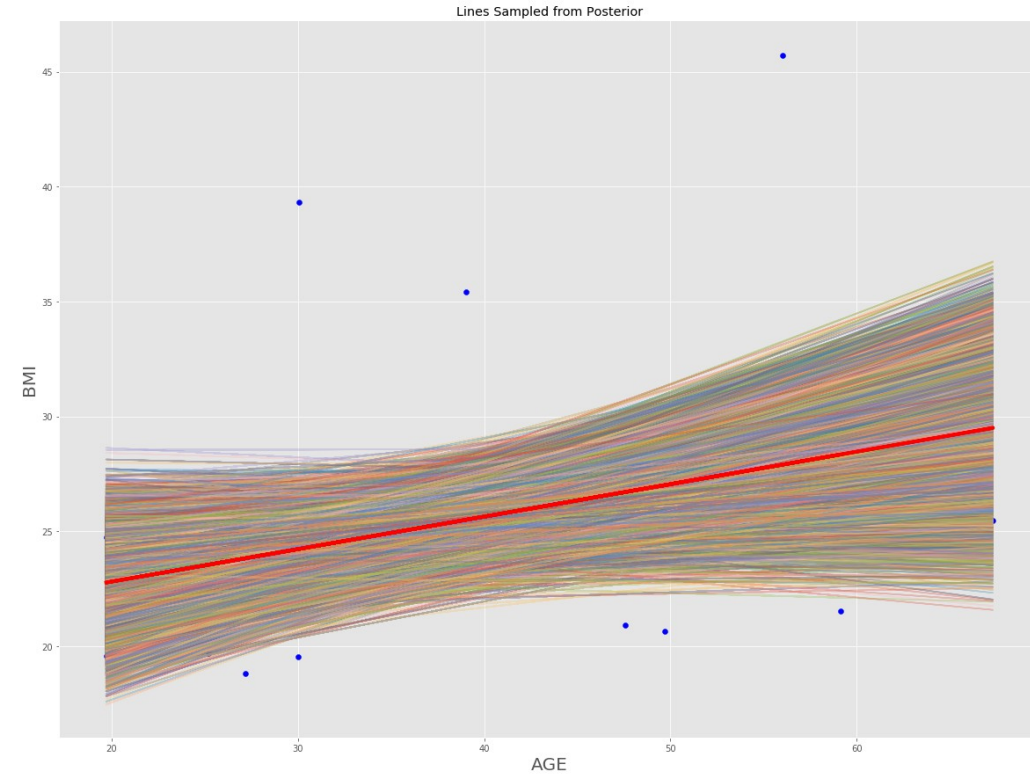
$N \approx P$



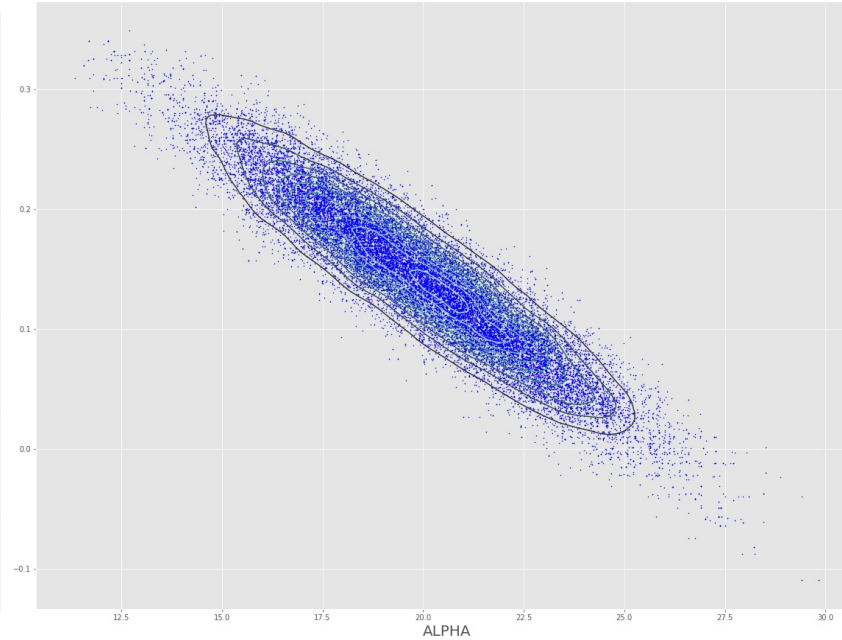
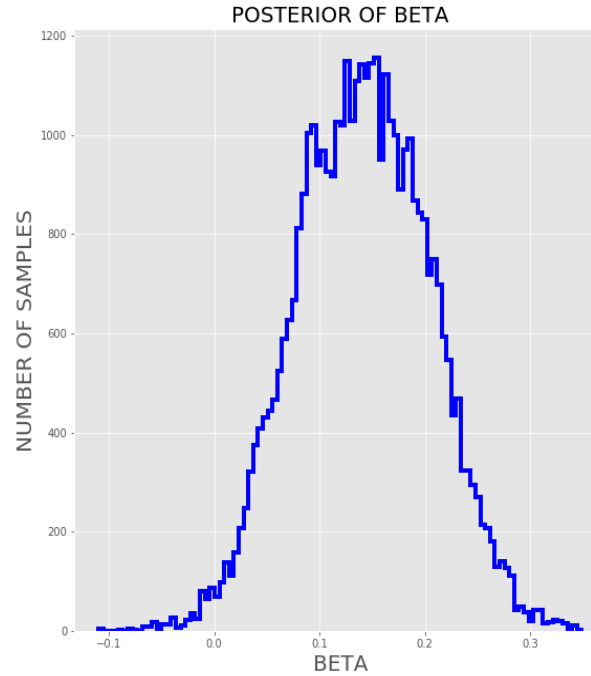
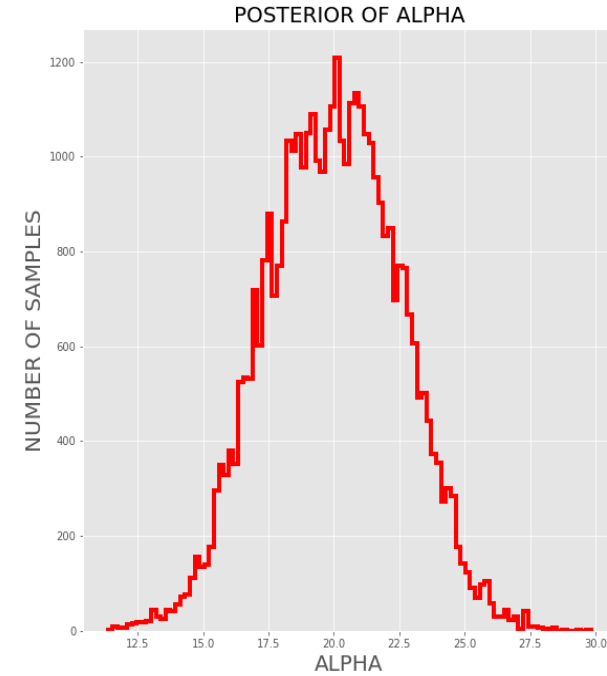
P is the number of features (genes, proteins, genetic variants etc.)
N is the number of observations (samples, cells, nucleotides etc.)



Fitting Frequentist linear model



Fitting Bayesian linear model



Slope and intersect are not fixed values any more

MCMC can draw infinite number of fits from proximity of optimal slope and intercept parameters

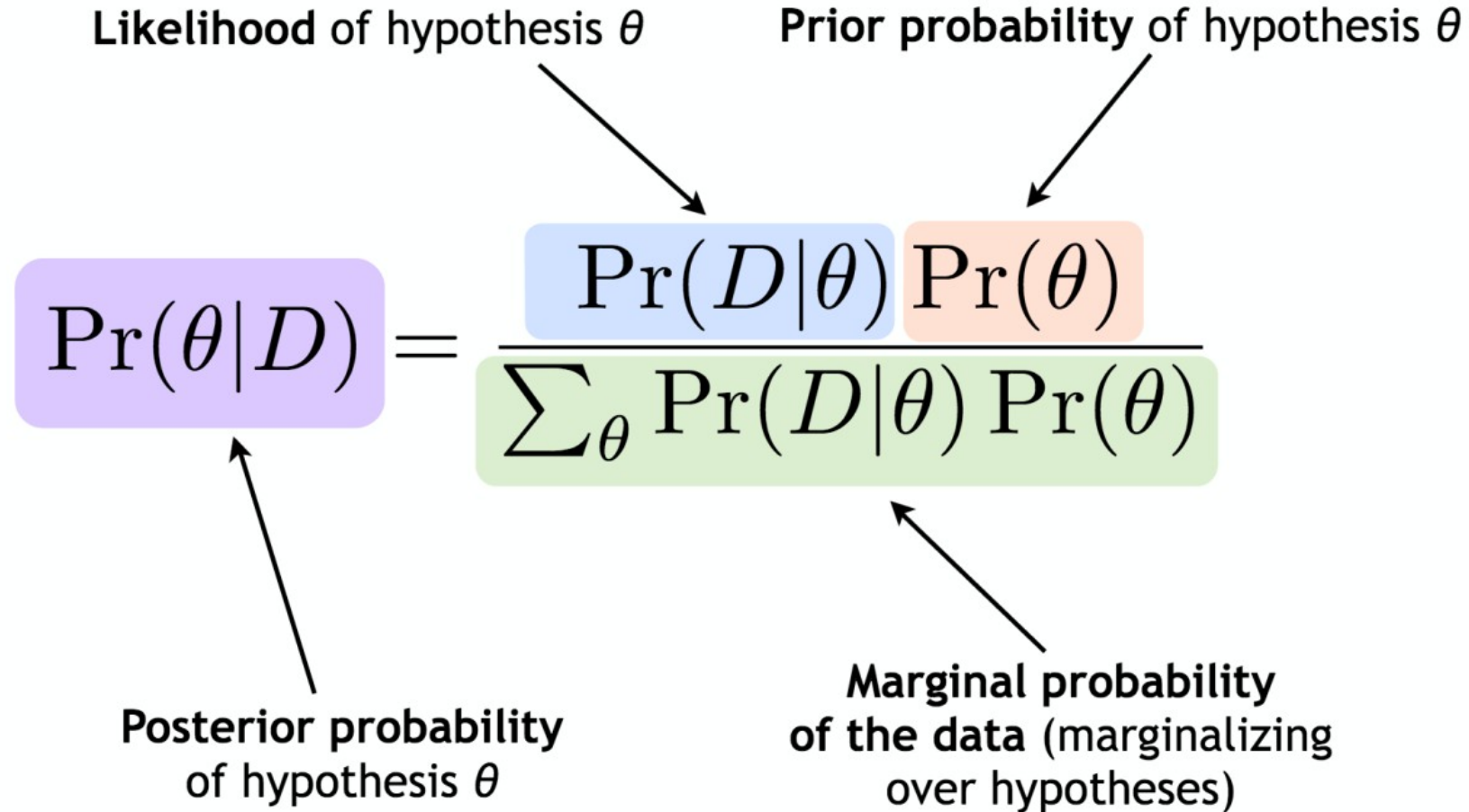
Likelihood of hypothesis θ

Prior probability of hypothesis θ

$$\Pr(\theta|D) = \frac{\Pr(D|\theta) \Pr(\theta)}{\sum_{\theta} \Pr(D|\theta) \Pr(\theta)}$$

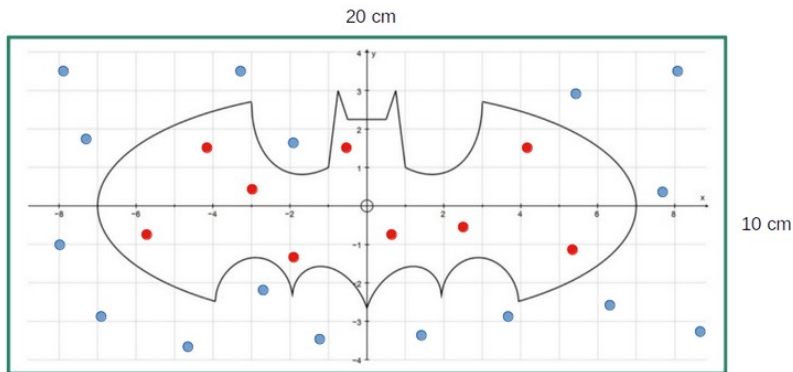
Posterior probability of hypothesis θ

Marginal probability of the data (marginalizing over hypotheses)



Markov Chain Monte Carlo (MCMC): introduction

- Integration via Monte Carlo sampling

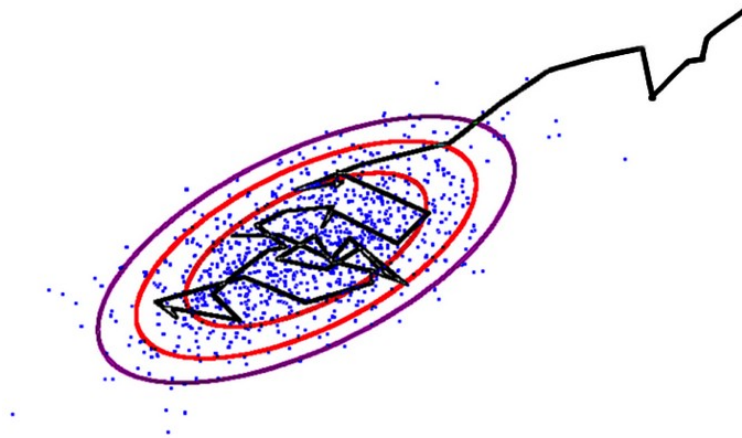


$$I = 2 \int_2^4 x dx = 2 \frac{x^2}{2} \Big|_2^4 = 16 - 4 = 12$$

```
1 f <- function(x){return(2*x)}; a <- 2; b <- 4; N <- 10000; count <- 0
2 x <- seq(from = a, to = b, by = (b-a) / N); y_max <- max(f(x))
3 for(i in 1:N)
4 {
5   x_sample <- runif(1, a, b); y_sample <- runif(1, 0, y_max)
6   if(y_sample <= f(x_sample)){count <- count + 1}
7 }
8 paste0("Integral by Monte Carlo: I = ", (count / N) * (b - a) * y_max)
```

[1] "Integral by Monte Carlo: I = 11.9248"

- Markov Chain Monte Carlo (MCMC)

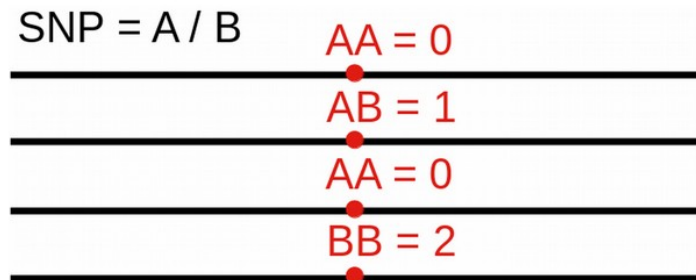


$$\text{Hastings ratio} = \frac{\text{Posterior}(\text{params}_{\text{next}} | \text{data})}{\text{Posterior}(\text{params}_{\text{previous}} | \text{data})}$$

- If Hastings ratio $> u$ $[0, 1]$, then accept, else reject
- Hastings ratio does not contain the intractable integral from Bayes theorem

Markov Chain Monte Carlo (MCMC) from scratch in R

- Example from population genetics



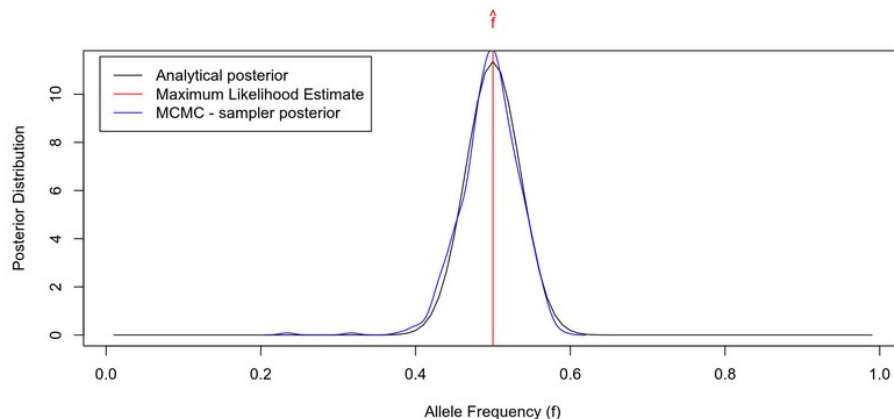
f – frequency of allele B; $g = 0, 1, 2$
 $n_g = n_0, n_1, n_2$ – genotype carriers

$$L(n | f) = \prod_g \left[\binom{2}{g} f^g (1-f)^{2-g} \right]^{n_g}$$

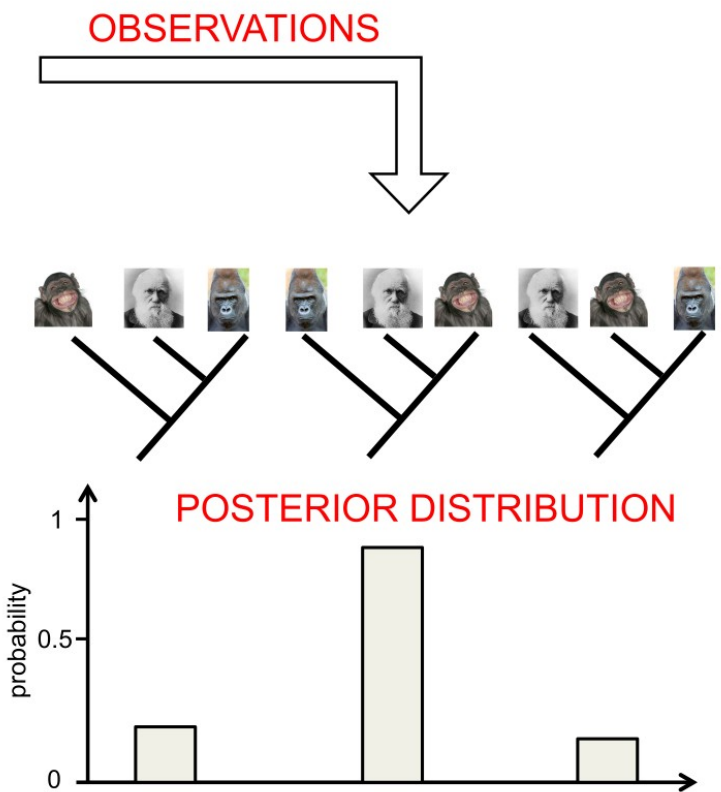
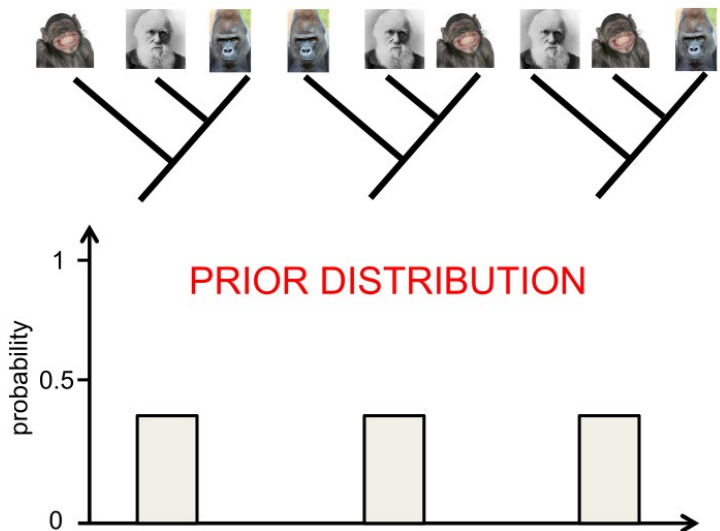
$$\frac{\partial \log[L(n|f)]}{\partial f} = 0 \Rightarrow \hat{f} = \frac{n_1 + 2n_2}{2(n_0 + n_1 + n_2)}$$

$$\text{Prior}(f, \alpha, \beta) = \frac{1}{B(\alpha, \beta)} f^{\alpha-1} (1-f)^{\beta-1}$$

```
1 N <- 100; n <- c(25, 50, 25) # Observed genotype data for N individuals
2 f_MLE <- (n[2] + 2*n[3]) / (2 * sum(n)) # MLE of allele frequency
3
4 # Define log-likelihood function (log-binomial distribution)
5 LL <- function(n, f){return((n[2] + 2*n[3])*log(f) + (n[2] + 2*n[1])*log(1-f))}
6 # Define log-prior function (log-beta distribution)
7 LP <- function(f, alpha, beta){return(dbeta(f, alpha, beta, log = TRUE))}
8
9 # Run MCMC Metropolis - Hastings sampler
10 f_poster <- vector(); alpha <- 0.5; beta <- 0.5; f_cur <- 0.1 # initialization
11 for(i in 1:1000)
12 {
13   f_next <- abs(rnorm(1, f_cur, 0.1)) # make random step for allele frequency
14
15   LL_cur <- LL(n, f_cur); LL_next <- LL(n, f_next)
16   LP_cur <- LP(f_cur, alpha, beta); LP_next <- LP(f_next, alpha, beta)
17   hasting_ratio <- LL_next + LP_next - LL_cur - LP_cur
18
19   if(hasting_ratio > log(runif(1))){f_cur <- f_next; f_poster[i] <- f_cur}
20 }
```



Combining priors with likelihoods



Thank you for your attention!

